Clinical Radiology 78 (2023) 83–98

Contents lists available at ScienceDirect

Clinical Radiology

journal homepage: www.clinicalradiologyonline.net

Review

Introduction to radiomics for a clinical audience

C. McCague[a,b,c,d,*], S. Ramlee[a], M. Reinius[b,c,d], I. Selby[a,d], D. Hulse[a,d], P. Piyatissa[a], V. Bura[a,d,e], M. Crispin-Ortuzar[b,c,f], E. Sala[a,b,d], R. Woitek[a,b,d,g]

[a] *Department of Radiology, University of Cambridge, Cambridge, UK*
[b] *Cancer Research UK Cambridge Centre, University of Cambridge, Cambridge, UK*
[c] *Cancer Research UK Cambridge Institute, University of Cambridge, Cambridge, UK*
[d] *Cambridge University Hospitals NHS Foundation Trust, Cambridge, UK*
[e] *Department of Radiology and Medical Imaging, County Clinical Emergency Hospital, Cluj-Napoca, Romania*
[f] *Department of Oncology, University of Cambridge, Cambridge, UK*
[g] *Research Centre for Medical Image Analysis and Artificial Intelligence (MIAAI), Department of Medicine, Faculty of Medicine and Dentistry, Danube Private University, Krems, Austria*

ARTICLE INFORMATION

Article history:
Received 2 August 2022
Accepted 31 August 2022

Radiomics is a rapidly developing field of research focused on the extraction of quantitative features from medical images, thus converting these digital images into minable, high-dimensional data, which offer unique biological information that can enhance our understanding of disease processes and provide clinical decision support. To date, most radiomics research has been focused on oncological applications; however, it is increasingly being used in a raft of other diseases. This review gives an overview of radiomics for a clinical audience, including the radiomics pipeline and the common pitfalls associated with each stage. Key studies in oncology are presented with a focus on both those that use radiomics analysis alone and those that integrate its use with other multimodal data streams. Importantly, clinical applications outside oncology are also presented. Finally, we conclude by offering a vision for radiomics research in the future, including how it might impact our practice as radiologists.

Introduction

Radiological imaging is a key component in the delivery of safe and efficient healthcare across the world. It is unique in offering spatially resolved, three-dimensional information about disease, which is acquired non-invasively and, upon serial imaging, allows for temporal comparison. Generally, radiological image interpretation is characterised by qualitative evaluation, which is often based on experience and prone to subjectivity,[1] or simple summary metrics such as lesion diameter or metabolic activity. Consequently, only a fraction of the wealth of quantitative information is extracted from medical images in clinical workflows.[2] The field of radiomics focuses on the extraction of quantitative features from digital images, which converts them into mineable high-dimensional data.[2] The driving hypothesis is that these features reflect the underlying biology but are imperceptible through the traditional visual inspection of current radiological practice.[3,4]

* Guarantor and correspondent: C. McCague, Department of Radiology, University of Cambridge, Cambridge, UK.
E-mail address: cm2074@medschl.cam.ac.uk (C. McCague).

https://doi.org/10.1016/j.crad.2022.08.149

The theory underpinning radiomics was first developed for the assessment of aerial photography[5] before its general application for a wide variety of image classification tasks.[6] Although early uses included exploring its potential in aiding the diagnosis of rheumatic heart disease[7] and classification of pulmonary disease on chest radiographs,[8] much of its more recent development has been focused in oncology[2] (see Figs 1 and 2).

Radiomics has developed as an offshoot of the wider "omics" fields in molecular biology. There, the advent of cost-efficient high-throughput techniques, advances in the fields of machine learning (ML) and artificial intelligence (AI), and computational power sufficient to process the downstream information has allowed extraction of huge volumes of highly complex data from multiple dimensions of disease molecular biology, including its DNA ("genomics"), RNA ("transcriptomics"), and associated proteins ("proteomics"), all of which can be extracted from a single biopsy.[2,9] Integrating these approaches allows a more holistic understanding of disease and helps to bridge the gap between genotype and phenotype.[9–11]

Radiomics, unlike the other "omics", is based on radiological imaging rather than invasive biopsy or molecular assays. The latter two may only capture disease at a specific anatomical site, or within a small portion of a tumour, are expensive to acquire, onerous for patients, and often impractical to take from more than one disease site or at serial time-points.[12] Radiomics analysis is performed on

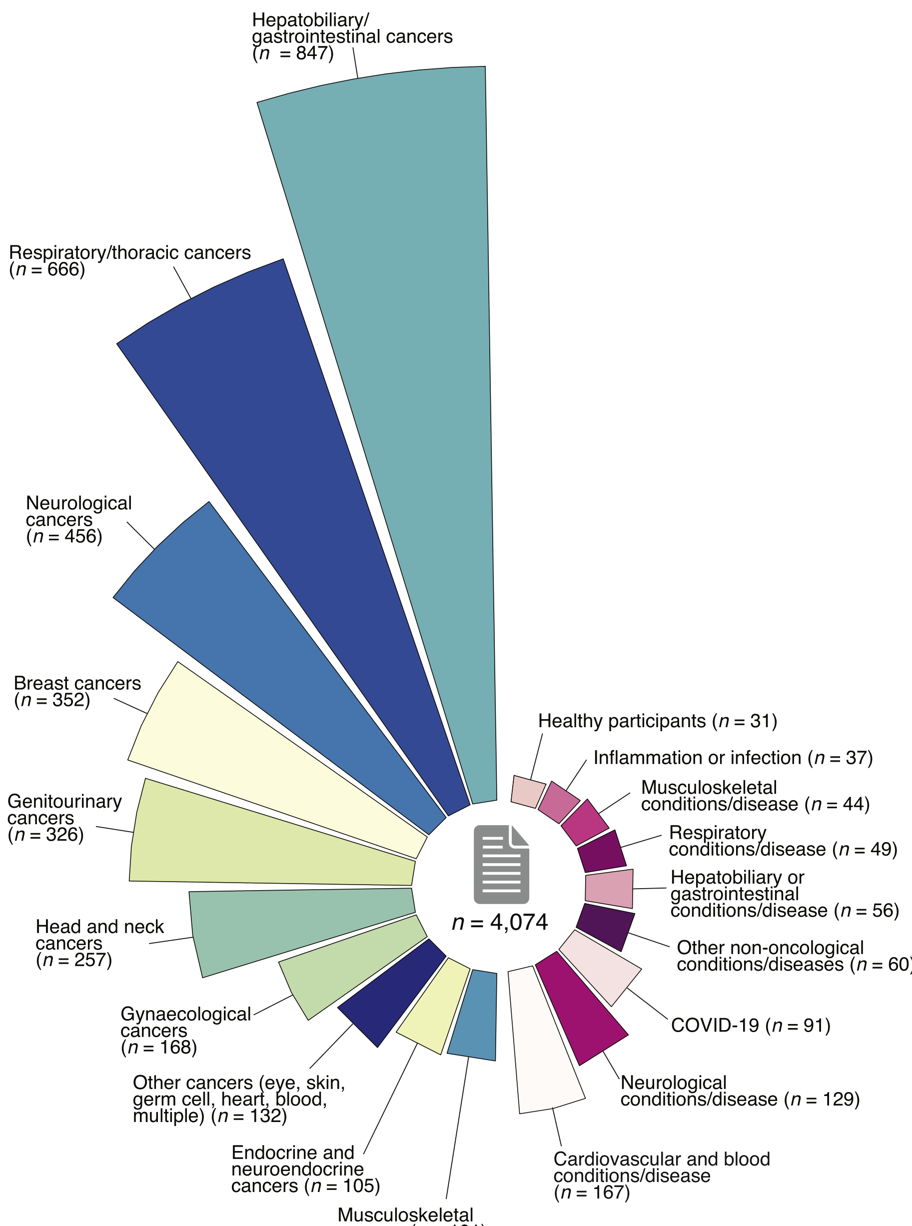

Figure 1 Circular bar chart showing the number of oncological and non-oncological radiomics studies published in PubMed. These exclude non-primary sources (reviews, proceedings, and errata), investigations on non-human models, and technical articles that do not interrogate a specific cancer site or condition/disease. At least 4,074 articles have been published as of 12 May 2022.

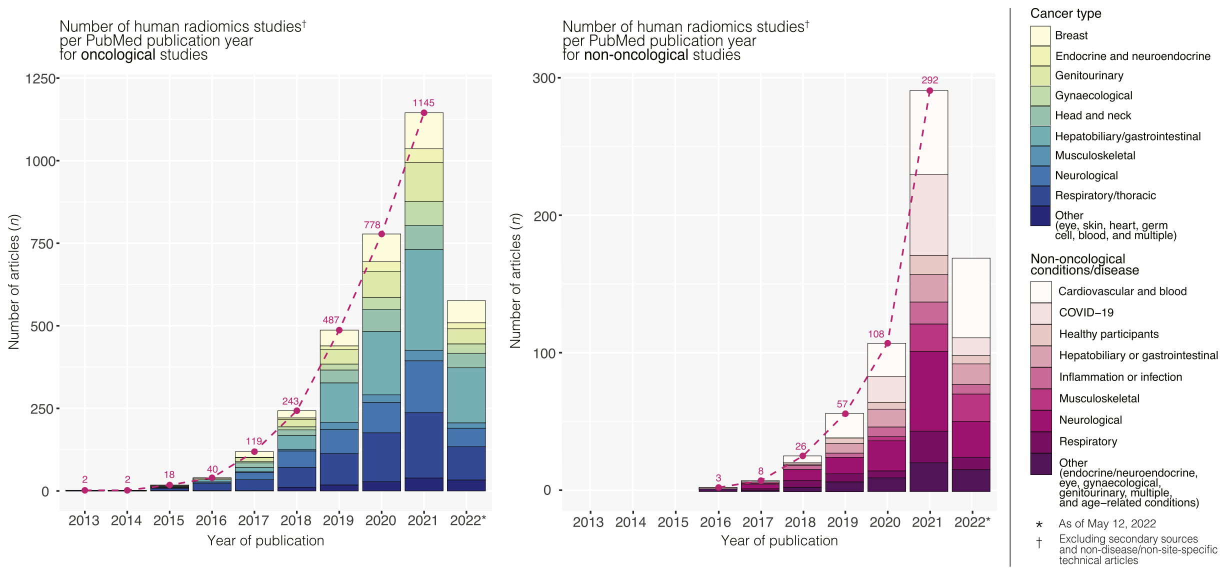

Figure 2 Bar charts showing the number of oncological- and non-oncological radiomics studies according to year of publication. After the word "radiomics" was coined in 2012 by Lambin *et al.*,[260] primary research articles investigating radiomics have materialised in the oncology space by 2013 and in other diseases by 2016—the number of articles for both areas have grown exponentially since then.

images acquired from standard imaging equipment at single or multiple time points during the course of a patient's routine clinical care, thus requiring no additional imaging to be performed and ideally covers the whole extent of a disease taking into account inter-lesional differences or heterogeneity.[1] The maximal utility of radiomics will likely be in its integration into other multimodal data. Models integrating radiomics signatures with genomic and clinicopathological information have shown improved prediction of disease outcome.[13–16]

This article aims to give an introduction to radiomics for a clinical audience, while highlighting recent developments in the field, and presenting a vision for future clinical applications of the technology, and how they will impact our practice as radiologists.

The process of radiomics extraction: more than meets the eye

This overview of the extraction process is focused on "traditional" radiomics analyses based on pre-set feature definitions (also called handcrafted features), as opposed to those based on deep learning (DL; see Fig 3 for overview of the radiomics pipeline). The process of radiomics analysis begins with the definition of the study aims and clinical question, as this will inform the optimal modality and protocol of medical imaging to be assessed, the regions of interest within the images, and ultimately the statistical analysis that will be performed.

Image acquisition

Radiomics analysis may be performed on images from multiple different techniques; however, work focusing on the use of computed tomography (CT) or magnetic resonance imaging (MRI) predominates within the published literature. Multiple intrinsic factors relating to image acquisition have potential to influence the radiomics signatures that are obtained, including hardware (scanner make and model), acquisition parameters (tube voltage and tube current for CT, field strength and pulse sequence for MRI, contrast medium administration for any technique) and reconstruction parameters (filters, voxel size), although the precise impact of such factors is the subject of ongoing debate and investigation.[17]

Segmentation

Before analysis can be performed, the images must undergo segmentation, which is the process of delineating the regions from which the features will be calculated. What constitutes the region of interest will depend on the aims of the study and could, for example, be a tumour on CT or MRI, or an area of opacity on radiographs.[18]

One of the key limitations of radiomics has been the time-consuming nature of manual image segmentation, performed slice-by-slice by an expert radiologist. Although this remains common, there is an increasing trend towards AI-based automated segmentation approaches, including fully or semi-automated segmentation (the latter with manual correction) using ML or neural network-based DL, although generalisability of such algorithms remains a limiting factor.[19]

Semi-automated segmentation has been shown to improve the repeatability and robustness of some radiomics features[20]; however, both manual[4,12] and semi-automated[4] segmentations carry the risk of considerable inter-reader variability, for instance due to the subjective assessment of lesions with ill-defined borders, which results in unstable radiomics features.[17] Therefore, ideally, multiple readers should segment at least a subset of the scans used for a radiomics study and some readers should also repeat their

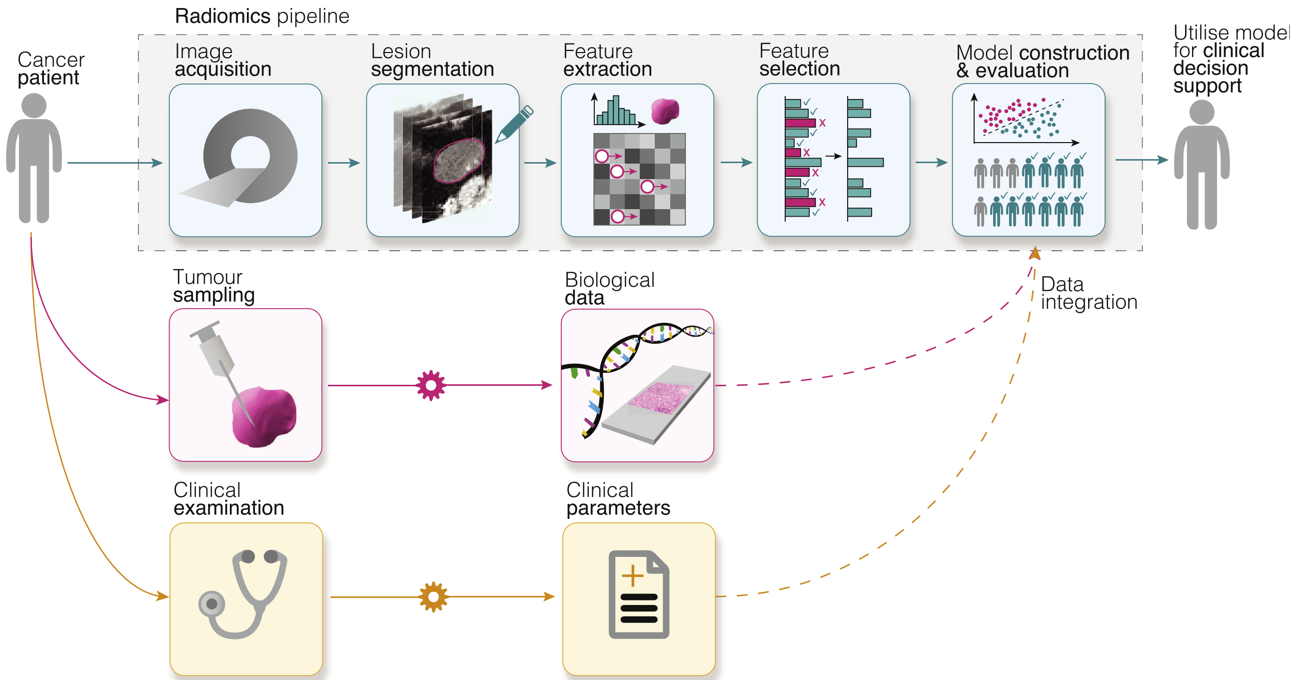

Figure 3 Schematic illustration of a typical radiomics pipeline. Radiological images acquired from medical scanners are first segmented to constrain radiomics analyses to the volume of interest (VOI), e.g., the tumour lesion. Radiomic features describing morphologic and textural characteristics are then extracted from VOIs. Because a large number of features tend to be computed, feature selection aims to identify which of these features are stable, non-redundant, and/or robust to any intrinsic dependencies (e.g., tumour volume). The result is a reduced set of only reliable radiomics features ("radiomics signature"), which are then used to construct, train, and evaluate mathematical models to predict outcomes (e.g., disease classification, treatment response, and survival). At this stage, non-imaging datasets, such as biological and clinical data, may also be integrated to develop holistic radiomics models that can be used as supporting tools for clinical decision-making.

segmentations after a period of time to assess the inter-reader variability and repeatability of radiomics. Focusing on radiomics features that show high repeatability and low inter-reader variability can be a very important step in the process of feature selection. AI-based automated segmentation is expected to partly alleviate the problem of inter-reader variability,[2,4,12] but compared to human readers they may be more sensitive to noise and artefacts[21]; however, it is still recommended that AI-based automated segmentations are reviewed and modified by a specialty reader if necessary.[2,12] Compared to human readers, AI-based segmenting tools can be less accurate due to high heterogeneity of the appearance of tumours, image noise, and artefacts.[21]

Multiple software packages are available for image segmentation and the extraction of radiomics features, including open-source and proprietary versions. Examples include 3D Slicer (3D Slicer community, https://www.slicer.org/), Microsoft advanced image labeller (project InnerEye, Microsoft, Redmond, WA, USA), and MIM software (MIM Software, Cleveland, OH, USA).

Image processing

Radiomics features extracted from a scan are sensitive to small changes in voxel intensity value and region of interest (ROI) size, which are dependent on the local imaging protocol, scanner make and model, and segmentation, which can affect the performance of a radiomics-based prediction model in different imaging conditions.[2,12,17,21,22] Therefore, following segmentation, a number of steps may be taken to harmonise the images from which features will be extracted. Harmonisation is a group of approaches that aim to produce consistent radiomics features from images acquired with different protocols.[3,17,23–25]

Many texture feature sets require isotropic voxel spacing to allow comparison between data from different cohorts. Range re-segmentation or intensity outlier filtering is performed to remove outlier voxels from the segmented region that fall outside of a specified range of grey levels.[18] Prior to feature calculation, the imaging data undergoes discretisation, a process by which individual voxel values are grouped into contiguous intervals (or "bins").[18,26]

Factors influencing the image produced, including imaging method, protocol, and relevant clinical conditions such as blood glucose in 2-[^{18}F]-fluoro-2-deoxy-D-glucose (^{18}F-FDG) positron-emission tomography (PET)[17] should be reported along with any harmonisation methods used, so that they can be considered when assessing and comparing studies.

Radiomics feature extraction

Medical image analysis has traditionally focused on qualitative assessment of visually appreciable "semantic" features, such as the size, shape, and contrast-enhancement of a lesion. Radiomics allows for a quantitative assessment of such features, and for the mathematical extraction of "agnostic" features that are not easily perceptible to the human eye, such as histogram features and texture analysis.[2] Shape analysis can include simple descriptions of size, such as dimensions; however, more complex topological characterisations may be extracted, such as roundness, compactness, spiculation, or convexity.[17] Texture features are the statistical interrelationships of the voxel intensities within the ROI.[27] First-order statistics (simple histogram features) relate to the properties of individual voxels without reference to their spatial distribution, such as mean, maximum, and standard deviation and also histogram skewness and kurtosis. Higher-order statistics, such as a grey-level co-occurrence matrix (GLCM) features (also known as Haralick features), grey-level run length matrix (GLRLM) features, and grey-level size zone matrix (GLSZM) features are also employed to assess the spatial relationship between voxels with certain intensity values systematically. Using predefined formulas, features describing these different matrices can then be calculated using standalone software or packages for integration into coding environments, such as Python or MATLAB (e.g., PyRadiomics, TexRAD, MaZda, IBEX, CERR).[28] Full lists of these so called "handcrafted" features including the standardised formulas to be used for their computation can be found on the International Biomarker Standardisation Initiative's (IBSI's) website.[29] Different filters may be applied to the original image to focus on a different frequency domain within the image, examples include wavelet and Gaussian filters.[30]

It has previously been reported that different software platforms could yield non-identical radiomics features calculations, calling into doubt the cross-platform reproducibility of the derived radiomics signatures.[23,31] Investigators are advised to conform to abovementioned guidelines published by the IBSI, and prospective investigators are encouraged to use IBSI-compliant software.[18]

Dimensionality reduction

Numerous features may be extracted, with provision for >100 features in many commonly used software packages increasing to >1,000 with the application of filters; however, only a relatively small proportion are likely to contribute significantly to a radiomics signature developed for the prediction of a clinically meaningful variable. Dimensionality reduction is a process by which the features are identified are least redundant, and this process contributes the most value to a generalisable radiomics model. This step is essential to avoid the issue of overfitting, which occurs when a model corresponds too closely to the training data and learns the noise within it rather than the underlying features relevant for robust predictions. Radiomics data sets are often relatively small with the number of extracted features exceeding the number of samples available for analysis, resulting in high dimensionality datasets particularly prone to overfitting.[32] Multiple strategies may be employed to reduce feature sets.[4,33,34] Approaches commonly include feature selection mechanisms such as ranking by mathematic criteria (intra-class correlation coefficients) or ranking features by importance metrics, and feature extraction methods in which the feature space is transformed to yield a new, smaller set of relevant features (e.g., linear discriminant analysis and principal component analysis [PCA]).

Model construction and evaluation

Both supervised and unsupervised learning can be applied in radiomics research. Unsupervised approaches do not necessarily require labelled training data and aim to uncover hidden structures, patterns, or sub-concepts in the data. Examples include clustering techniques and PCA.[35] It can then be determined whether these patterns have an association with the desired outcome.[36]

In a supervised approach, the radiomics model is fitted to the target variable or study endpoint, for example, tumour histology or disease progression, and will typically incorporate the most strongly predictive radiomics features.[36] Least absolute shrinkage and selection operator (LASSO) is a very popular supervised ML method in which the process of feature selection happens in tandem with the model construction (a regression analysis is applied on the training data and features with a coefficient of zero are removed).[35] In contrast, feature selection and model construction happen separately in recursive feature elimination support vector machines (SVM-RFE), where the model needs to be rebuilt again after the selection of the features.[37]

Supervised ML models are trained on a subset of the available images called the "training" data. Then, the performance and generalisability of the model is evaluated by validation testing. Two types of validation approach exist. The first approach, internal validation, makes use of similar data from which the model was derived. Popular examples of this approach are the holdout, and cross-validation methods. In the hold-out method the original dataset is split in two, with the model developed on the training subset, and then tested on the remainder of the data. For smaller sample sizes, cross validation is typically performed, where the data are split into several "folds" and then used sequentially to train and to validate the model.[38] The second approach is called external validation, where the data used to test model performance are structurally different to the data used to train the model and have usually been acquired from a separate institution in a different region.[39]

The risk of bias in the model can be reduced by using large enough data sets obtained across multiple centres and populations; however, a perennial issue within the field of radiomics is that of small sample sizes, often comprising only a few hundred cases.[2] There are several possible reasons for this, including the labour-intensive nature of manual segmentation, and restricted access to external data.[38]

Guidelines for model development

Feature extraction and selection

Different software and calculation methods are used to extract radiomics features from a scan. The IBSI is designed to standardise calculation methods,[18] but there is still variation in the quantitative output based on platform and software version, which may impact predictive power.[23] IBSI compliance, feature extraction, software type and version are therefore important to report.[23]

With a higher number of features, there is an increasing chance that some irrelevant features happen to correlate with the outcome in the sample population, creating a danger of over-fitting the radiomics model to the sample, instead of to the population as a whole.[2,3,12,24,40] Validation helps identify overfitting,[3] as an over-fitted model would perform well on the test dataset but poorly on new data.[3,24] Validation should ideally be performed on an external dataset to provide a more realistic estimate of the model's performance[12,17]; however, external validation is as rare as 6% in published literature.[41] Options to limit over-fitting include using fewer radiomics features, such as in hand-crafted radiomics,[40] using a larger sample size,[12] or reducing the number of radiomics features assessed by the model using unsupervised feature selection.[3,12]

Evaluating the quality of radiomics publications

Performance metrics should include sensitivity and specificity in addition to summary metrics such as accuracy, to ensure that possible training biases are adequately captured. The importance of including such detail is highlighted in cases of class imbalance, a situation where most of the data present belongs to only one class label, such as disease presence for example. A model trained to predict a disease that affects 1 in 100 people might achieve 99% accuracy by labelling all possible patients as disease free, but yields 0% sensitivity, and thus have no practical clinical utility.[3]

Published radiomics research is of variable quality regarding methodology, validation, and reporting of results.[4] There are tools available to assess the quality of a publication, such as the Radiomics Quality Score (RQS). Scored out of 36, it includes points for study design, statistical analysis, reporting of results, and open access to the tools and data used.[42] Previous studies suggest that the current standard of published literature is insufficient based on RQS.[4] The RQS itself, however, is not a comprehensive assessment of quality, as, for example, it does not include points for harmonisation techniques.[3,43]

DL methods can bypass the need for segmentation and feature extraction, and instead assess the prognostic value of each voxel or area in a scan[21] but are not the focus of this review. They come with their own inherent pitfalls, where the AI tool may draw spurious associations to the prognosis that are difficult to check as the results of DL tools can be difficult to understand or interpret biologically.[24] Some pitfalls of DL methods not captured by the RQS are included in the Checklist for AI in Medical Imaging (CLAIM), for example, points for explainability or interpretability using saliency maps[44] (see Fig 4 for further information on DL-based approaches, and saliency maps). Other tools also exist to both help design and evaluate research, such as transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD), prediction model risk of bias assessment tool (PROBAST), their extensions for AI and ML tools TRIPOD-AI and PROBAST-AI (both currently in development), and Quality Assessment of Diagnostic Accuracy Studies (QUADAS).[45,46]

Clinical applications of radiomics

Oncological applications of radiomics

As illustrated in Figs 1 and 2, the field of radiomics research has, to date, been dominated by studies in oncology. Radiomics tools have been used in studies related to screening, disease detection, diagnosis, staging and prognosis, finding and predicting biological correlates, prediction of treatment response, and those that further our understanding of key oncogenic processes.[47] This section will highlight studies related to the prediction of biological correlates, stage and treatment response, in addition to studies that integrate radiomics and other multi-modal data streams.

Predicting histological or molecular features of tumours, and their stage

Accurate biological subtyping is paramount to the selection of oncological treatment, but can be limited by access of sufficient diagnostic tissue and tumour heterogeneity. The use of radiomics in predicting key histotypes has been researched extensively to address this challenge in a variety of tumour types: these have included oestrogen receptor (ER)/progesterone receptor (PR)/human epidermal growth factor receptor 2 (HER2) status prediction from MRI, PET/CT, mammography and low-dose CT in breast cancer,[48,49] and the prediction of small-cell or non-small-cell adenocarcinoma/squamous and large-cell pathologies from CT or PET/CT at lung cancer diagnosis.[50–52] Radiomics predictors of microvascular invasion, a histopathological predictor of poor recurrence-free and overall survival in surgically treated hepatocellular carcinoma (HCC), have also been reported.[53–55]

With regard to studies aiming to predict the mutational status of specific genes, published work to date has focused predominantly on established biomarkers currently applied in cancer therapy. These include abnormalities affecting EGFR/ALK, BRCA, KRAS/BRAF/NRAS, IDH/1p19q (co-deletion), and VHL/BAP1/PBRM1 in lung, breast, colorectal, central nervous system (CNS), and renal cancers, respectively (as reviewed in[56]). Gene expression-based biomarkers predicted by radiomics have included OncotypeDX in breast cancer,[57,58] as well as key prognostic transcriptomic subtypes including in CNS,[59–61] hepatobiliary,[62] lung[63] and gynaecological[64] tumours. Radiomics-based DNA or RNA methylation status prediction has also been reported, such as for MGMT promoter methylation in glioblastoma,[65–67] m6A RNA methylation in urothelial tumours,[68] and methylation-based subgrouping in renal cell carcinoma[69] and squamous cell carcinomas of the oral cavity, larynx, or pharynx.[70]

With the rising prominence of immune checkpoint inhibitors (ICIs) in oncology, radiomics research to predict molecular biomarkers of ICI response has expanded in

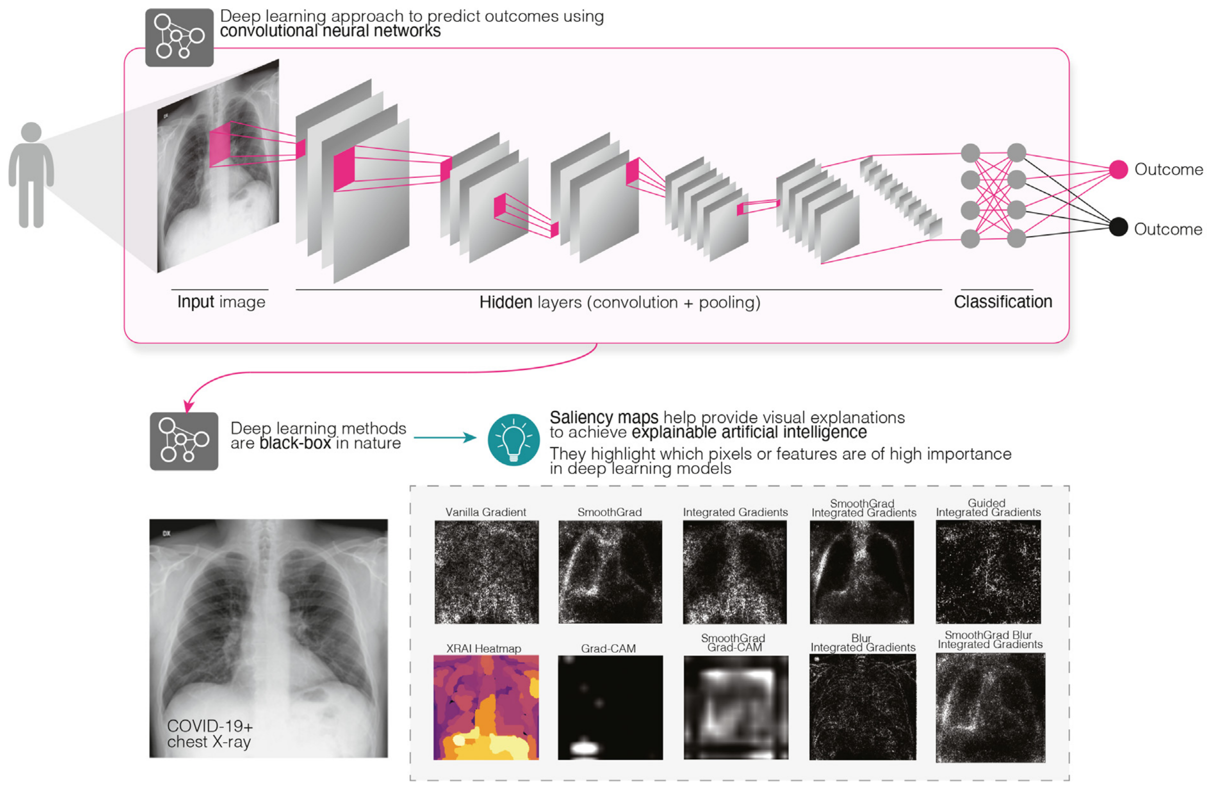

Figure 4 DL approaches are black box in nature. Convolutional neural networks (CNNs) contain many hidden and complex layers. What is learned in these layers can be enigmatic and often means that results are difficult to interpret. To achieve explainable AI, saliency maps help visualise which features on the input images are particularly important while a CNN is making a prediction. In other words, these maps tell us which parts of the image are paid most attention to for a CNN to derive an outcome. Here, we show a variety of saliency maps computed in Python v3.8.5 (Python Software Foundation, Wilmington, Delaware, USA) with TensorFlow CNNs (v2.9.1)[261] and the Saliency library (v0.2.0).[262] The sample chest radiograph was obtained from an online COVID-19 database.[263,264]

recent years. The presence of tumour-infiltrating lymphocytes (TILs) has been strongly associated with favourable oncological outcome[71] and ICI response,[72] and the TIL–radiomics relationship has been explored most extensively in breast cancer using MRI or mammography images.[48,73–76] Radiomics prediction of CD8-positive T-lymphocyte presence or abundance has been demonstrated in a number of tumour-type specific studies covering hepatobiliary,[77,78] lung,[79] and GI[80,81] cancers, as well as in a pancancer study, which further achieved predictive significance in terms of ICI response in a subset of selected tumour types.[82] Other ICI biomarkers predicted radiomically in the literature include measures of tumour mutational burden,[83,84] microsatellite instability[83,85–87] and PD-1/PD-L1/PD-L2 expression.[88–90]

Preoperative staging is a further area that may benefit from the use of radiomics. MRI radiomics can predict head and neck cancer stage,[91] while CT-based radiomics showed significant associations with overall stage and primary tumour stage in lung cancer.[92] In addition, in both lung and head and neck cancer patients, the addition of radiomics improved prognostication based on TNM stage alone.[92] Several other studies showed good prediction of metastatic disease in lung cancer patients using CT[93,94] or PET/CT[95] radiomics. In addition, diffusion-weighted imaging (DWI) radiomics has been shown to predict T, N, and overall stage, as well as perineural invasion in gastric cancer patients.[96–98]

Predicting response to treatment

Exploration of the clinical utility of radiomics has expanded in recent years to include predicting response to specific oncological treatments. Prediction of the pathological and/or radiological response to neoadjuvant chemotherapy or chemoradiotherapy using MRI radiomics has been explored extensively in breast[99] and rectal[100–104] cancers, and has also been reported in cervical[105] and head and neck[106] tumours. CT radiomics has been utilised in predicting response to first-line chemotherapy in gastric,[107] bladder,[108] lung,[109,110] and ovarian[15,111,112] tumours, and to neoadjuvant chemoradiotherapy response in oesophageal[76,113] and lung[114,115] cancers. Other oncological therapy classes for which responses have been predicted using radiomics have included anti-EGFR therapies in lung cancer,[116–118] anti-HER2 treatment in breast cancer,[119,120] transarterial chemoembolisation in HCC,[121,122] and ICIs in a number of tumour types.[82,123,124] Importantly, multi-stream data integration has been shown to improve radiomics-based neoadjuvant response prediction.[15,125–127]

In addition, the field of delta-radiomics has become popular in recent years in assessing treatment response.

Delta-radiomics measures changes in imaging features over time, usually before and after therapy. This can reveal subtle early tumoural changes that precede measurable changes in size, thus having the potential to aid as a complementary tool to RECIST assessment. The use of delta-radiomics in treatment response assessment in brain, head and neck, lung, gastro-intestinal, colorectal, breast, prostate, kidney, and other malignancies (sarcoma, melanoma, and haematological) has been covered in a recent systematic review.[128] With the increasing use of immunotherapy in oncology, recent studies utilising delta-radiomics analyses have shown good prediction of immunotherapy treatment response[129–134] as well as the potential to discriminate between pseudo-progression and true progression early during treatment.[130,135]

Integrating radiomics with other data streams

The relationship between imaging features and associated tumour biology is under active investigation in the rapidly expanding field of *radiogenomics*. This integration of radiomics with other data types, including clinical and molecular data, has been shown in a range of tumour types to improve the accuracy of survival prediction compared to single modality predictions alone. Clinical data, such as nodal status,[13,136] luminal/HER2-positive/triple-negative subtype,[13] histological proliferation index or tumour size[136] have contributed to integrated MRI radiomics-based nomogram approaches that allow superior prediction of disease-free survival compared to imaging or clinical data alone in breast cancer. MRI radiomics features were shown to have superior prognostic significance compared to clinical and radiological models alone in glioblastoma, but performed best when integrated with clinical data including extent of surgical debulking, patient age, and performance status.[137]

The addition of prognostic molecular markers to clinical and radiomics data has been associated with incremental enhancement in predictive performance, for instance by integrating IDH mutational[14,138,139] or MGMT promoter methylation[14,138–141] status with MRI radiomics and clinical features in glioblastoma. Clinical measures of routine circulating markers can also contribute to improved outcome prediction using radiomics, for instance with the inclusion of prostate-specific antigen (PSA)[142] and carcinoembryonic antigen (CEA)[143] levels within MRI- and CT-based radiomics–clinical modelling of prostate and pancreatic cancers, respectively; however, in patients with ovarian cancer, the prediction of response to neoadjuvant chemotherapy based on radiomics, clinical data, and CA125 could not be improved significantly when circulating tumour DNA (ctDNA) assessment was included.[15] The integration of gene expression data with CT, MRI, or PET radiomics has also been shown to increase the accuracy of survival predictions across various tumour types.[144–147]

Non-oncological applications of radiomics

Radiomics is potentially applicable to many diseases, with the only requirement being a dataset containing a radiological volume-of-interest and an outcome. Support coming from cancer imaging programmes in particular[148,149] and the greater availability of oncological datasets has resulted in other applications being less common and well-established.[2,150] Although much of the radiomics research outside oncology is in its infancy, there is growing interest in its utility in a variety of other diseases (see Fig 2), with promising results in several areas (see Fig 5 for overview of non-oncological applications of radiomics).

Neurological

Diagnosis and classification of both acute ischaemic and haemorrhagic stroke have been demonstrated,[151–154] whilst multiple prognostic applications also show potential. Functional outcome at discharge,[155–158] evaluation of cognitive impairment,[159] risk of haemorrhagic transformation of ischaemia,[160] risk of haematoma expansion,[161–164] and risk of malignant cerebral oedema[165–167] have all been investigated. CT/CT angiography (CTA) radiomics may indicate the likelihood of recanalisation following administration of thrombolysis[158,168] or mechanical thrombectomy,[169,170] which could ultimately prevent patients undergoing high-risk, but likely futile, treatment.

Textural analysis of multiple sclerosis (MS) was first explored in 1999 when Mathias *et al.* analysed the potential for monitoring spinal cord lesions.[171] Greater textural heterogeneity on MRI has been shown to be a potential measure of tissue integrity, correlating with more severe myelin and axonal pathology.[172] Multiple studies have subsequently demonstrated that active lesions can be distinguished from chronic, indolent plaques[173–175] and may allow for early diagnosis.[176] Other neurological applications include Parkinson's disease,[177–182] Alzheimer's disease[183,184] and schizophrenia.[185–187]

Cardiovascular

Radiomics applied to coronary atherosclerotic plaques has been shown to be predictive of major cardiac events and may distinguish advanced from early plaques.[188–190] It can even identify unique morphologic patterns specific to specific cardiovascular risk factors, e.g., cocaine-use,[191] demonstrating the potential for individualised management.[192]

Cardiac CTA radiomics may allow for accurate assessment of the myocardium where cardiac MRI is not available and reduce or even avoid the use of gadolinium.[193] Multiple studies suggest it is possible to diagnose and subclassify acute[194–200] and chronic[201–203] myocardial infarction, myocarditis,[204,205] and cardiomyopathy[19,206–209] on cross-sectional imaging. Applying these techniques to echocardiography may allow for detection of left ventricular remodelling[210] and may be able to differentiate pannus from thrombus or vegetation in patients with suspected prosthetic valve obstruction on CT.[211]

Studies have additionally demonstrated potential applications to vascular disease, for example in predicting the risk of expansion and rupture of abdominal aortic aneurysms[212] and outcome after endovascular aortic repair (EVAR),[213,214] whilst textural analysis of carotid ultrasound images has been shown to determine stroke risk from carotid artery disease.[215,216]

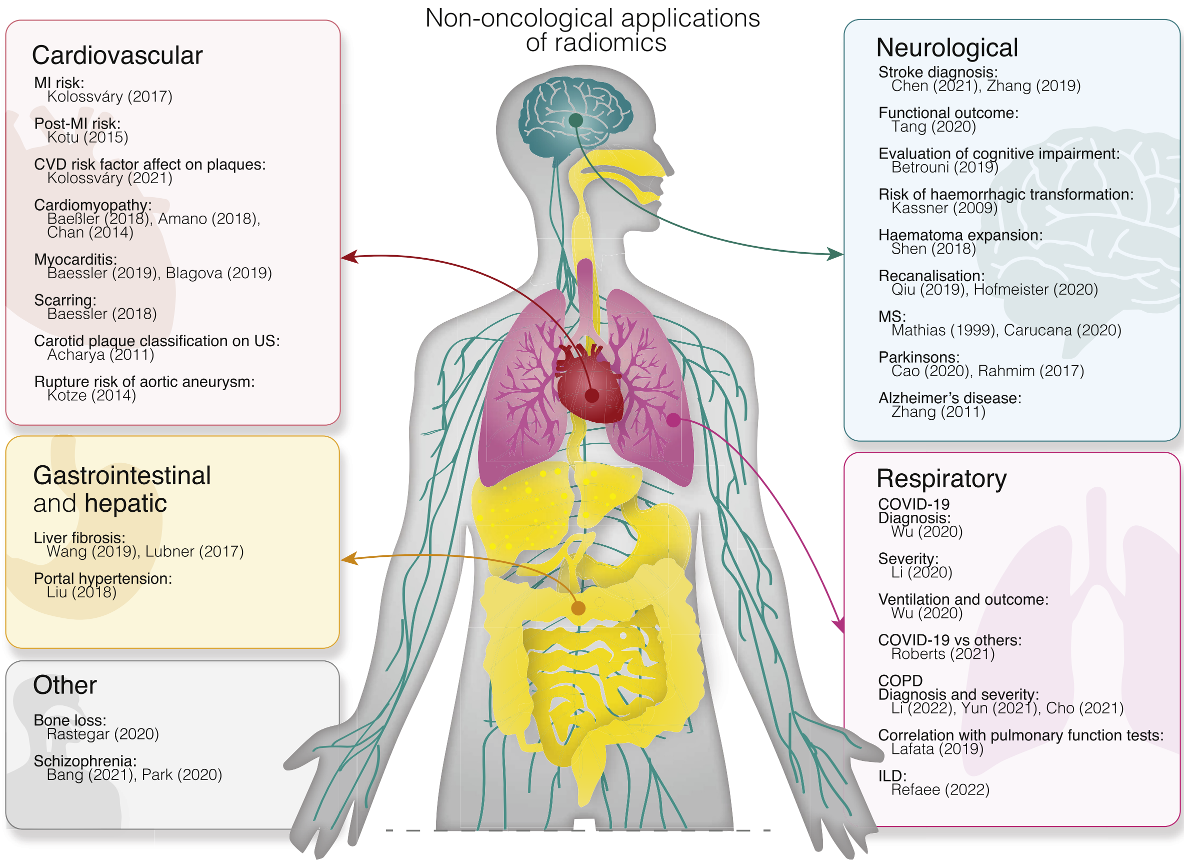

Figure 5 Non-oncological applications of radiomics.

Respiratory

Since the onset of the coronavirus disease 2019 (COVID-19) pandemic, there have been a large number of studies examining potential uses of radiomics. Attempts have been made to diagnose[217–219] and monitor[220] the disease, predict severity[221–223] or the need for mechanical ventilation or mortality[224–226] using chest radiography and CT. The ability to distinguish COVID-19 from influenza and other non-COVID pneumonias may also be possible[227–231]; however, the standards of research have failed to allow for meaningful clinical adoption during the pandemic to date.[45]

Diagnosis, severity, and prognosis of chronic obstructive pulmonary disease (COPD) may be predicted by radiomics analysis,[232–235] and results correspond to pulmonary function tests.[232,236,237] It may be possible to stratify risk of developing the condition[233,238] and identify sub-clinical smoking-related lung injury.[239] Ultimately, it may allow for more accurate identification of disease phenotypes, allowing for more personalised and targeted therapies.[240–242]

Similar applications may be found in interstitial lung disease (ILD), where studies have attempted to diagnose idiopathic pulmonary fibrosis (IPF),[243] discriminate between multiple types of ILD,[244] and quantify risk of lung cancer.[245]

Gastrointestinal and hepatic

A prospective, multicentre study found that radiomics assessment of elastography was an excellent surrogate of histological staging in hepatic fibrosis and is superior to conventional two-dimensional shear-wave elastography.[246] Radiomics of contrast-enhanced CT may provide similar insight, with subtle changes in heterogeneity corresponding to the presence and severity of fibrosis.[247,248] Furthermore, radiomics may offer a non-invasive opportunity to diagnose clinically significant portal hypertension.[249]

Radiomics may be able to assist in differentiating Crohn's disease from ulcerative colitis.[250] In Crohn's disease, it may be possible to identify patients at higher risk of surgery[251] or loss of response to biological treatments,[252] assisting clinicians to identify patients who require stricter monitoring or different therapeutic regimes.

Non-oncological integrating radiomics with other data streams

Although non-oncological multiomics integration has been limited to date, radiogenomics has shown promise in identifying patients with mild cognitive impairment who are most at risk of Alzheimer's disease, potentially allowing for personalised treatment aiming to slow progression.[253–255] In IPF, a radiogenomics approach provided additional insight into disease progression and response to treatment.[256]

Radiotranscriptomics applied to peri-vascular adipose tissue on CT may be a marker of tissue inflammation and has potential prognostic applications in coronary artery disease[257] and COVID-19.[258] Furthermore, this approach suggested the alpha variant of COVID-19 caused higher vascular inflammation than the original strain and treating such patients with dexamethasone improved outcomes.[258]

Conclusion

The Royal College of Radiologists believe the advent of medicine's AI age will herald the most fundamental change in healthcare delivery since the foundation of the NHS over 70 years ago.[259] Of these AI-associated health technologies, radiomics may prove the most impactful by allowing us to bridge the gap between medical imaging and personalised medicine.[42]

To date, research has been focused on oncological applications; however, this review has highlighted the increasing use of radiomics techniques in a raft of other diseases.

In order to create tools with clinical utility, prospective trials that validate radiomics signatures on external datasets are required. Therefore, safe sharing of datasets between centres or publicly is a requirement for the field to develop further and for radiomics-based AI tools to make the transition into clinical workflows. There is also need for standardisation of analysis, in line with those recommended by IBSI. Although standardisation of image acquisition would be beneficial, it is unlikely to be feasible between different vendors and scanners. Therefore, research on the identification of predictive radiomics that remain robust, especially against differences in image acquisition and reconstruction, is more important than ever. In addition, heterogeneous datasets for model training and testing, and entirely independent data sets for validation, would be desirable for all radiomics studies. For the clinical setting, an interdisciplinary approach combining the knowledge and insight of key stakeholders, including radiologists, pathologists, oncologists, medical physicists, imaging scientists, data scientists, and patients, is necessary to accelerate the penetrance of radiomics-based tools into patient care.

As the domain experts in medical imaging, radiologists are uniquely placed to lead these developments, which will augment our diagnostic skills. This will require radiologists to educate themselves on the basics of how tools based on radiomics and AI are designed and tested, so that we can play an active role in their development, and appraise and scrutinise their utility on our patients' behalf. Given the prospective centrality of these tools in the provision of healthcare, changes to both undergraduate medical and radiology training curriculums may be warranted. These efforts would reinforce the value of radiology into the future.

Declaration of competing interest

The authors declare no conflict of interest.

Acknowledgements

The funders for the project are: NIHR BRC Cambridge, United Kingdom; The Mark Foundation for Cancer Research, United States; Cancer Research UK, United Kingdom; Innovative Medicines Initiative, Belgium; EPSRC, United Kingdom; Horizon 2020, European Union; CRUK, United Kingdom; EFPIA, Belgium; The Austrian Science Fund, Austria; Wellcome Trust, United Kingdom. The views expressed are those of the authors and not necessarily those of the NHS, the NIHR, or the Department of Health and Social Care.

References

1. Hosny A, Parmar C, Quackenbush J, *et al.* Artificial intelligence in radiology. *Nat Rev Cancer* 2018;**18**(8):500–10.
2. Gillies RJ, Kinahan PE, Hricak H. Radiomics: images are more than pictures, they are data. *Radiology* 2016;**278**(2):563–77.
3. Mayerhoefer ME, Materka A, Langs G, *et al.* Introduction to radiomics. *J Nucl Med* 2020;**61**(4):488–95.
4. van Timmeren JE, Cester D, Tanadini-Lang S, *et al.* Radiomics in medical imaging—"how-to" guide and critical reflection. *Insights Imaging* 2020;**11**(1):1–16.
5. Kassner A, Thornhill R. Texture analysis: a review of neurologic MR imaging applications. *AJR Am J Neuroradiol* 2010;**31**(5):809–16.
6. Haralick RM, Shanmugam K, Dinstein IH. Textural features for image classification. In: *IEEE transactions on systems, man, and cybernetics*vol. 3; Nov. 1973. p. 610–21. https://doi.org/10.1109/TSMC.1973.4309314.
7. Hall EL, Kruger RP, Dwyer SJ, *et al.* A survey of preprocessing and feature extraction techniques for radiographic images. *IEEE Trans Comp* 1971;**100**(9):1032–44.
8. Sutton RN, Hall EL. Texture measures for automatic classification of pulmonary disease. *IEEE Trans Comp* 1972;**100**(7):667–76.
9. Subramanian I, Verma S, Kumar S, *et al.* Multi-omics data integration, interpretation, and its application. *Bioinform Biol Insights* 2020;**14**:1177932219899051.
10. Hasin Y, Seldin M, Lusis A. Multi-omics approaches to disease. *Genome Biol* 2017;**18**(1):1–15.
11. McCague C, Beer L. Radioproteomics in patients with ovarian cancer. *Br J Radiol*;94(1125):20201331.
12. Papanikolaou N, Matos C, Koh DM. How to develop a meaningful radiomic signature for clinical use in oncologic patients. *Cancer Imaging* 2020;**20**(1):1–10.
13. Park H, Lim Y, Ko ES, *et al.* Radiomics signature on magnetic resonance imaging: association with disease-free survival in patients with invasive breast cancer. *Clin Cancer Res* 2018;**24**(19):4705–14.
14. Beig N, Bera K, Prasanna P, *et al.* Radiogenomic-based survival risk stratification of tumor habitat on Gd-T1w MRI is associated with biological processes in glioblastoma. *Clin Cancer Res* 2020;**26**(8):1866–76.
15. Crispin-Ortuzar M, Woitek R, Bura V, *et al.* Integrated radiogenomics models predict response to neoadjuvant chemotherapy in high grade serous ovarian cancer. *medRxiv* 2021;**7**(22):21260982, https://doi.org/10.1101/2021.07.22.2126098.
16. Yang J, Chen J, Kuang K, *et al.* MIA-prognosis: a deep learning framework to predict therapy response. In: *Medical image computing and computer assisted intervention – MICCAI 2020. 23rd international conference, Lima, Peru, October 4–8, 2020, Proceedings, Part II*. New York: Springer International Publishing; 2020.
17. Rizzo S, Botta F, Raimondi S, *et al.* Radiomics: the facts and the challenges of image analysis. *Eur Radiol* 2018;**2**(1):1–8.
18. Zwanenburg A, Vallières M, Abdalah MA, *et al.* The image biomarker standardization initiative: standardized quantitative radiomics for high-throughput image-based phenotyping. *Radiology* 2020;**295**(2):328–38.
19. Baeßler B, Mannil M, Maintz D, *et al.* Texture analysis and machine learning of non-contrast T1-weighted MR images in patients with hypertrophic cardiomyopathy—preliminary results. *Eur J Radiol* 2018;**102**:61–7.
20. Haniff NSM, Abdul Karim MK, Osman NH, *et al.* Stability and reproducibility of radiomic features based various segmentation technique on MR images of hepatocellular carcinoma (HCC). *Diagnostics* 2021;**11**(9):1573.
21. Corrias G, Micheletti G, Barberini L, *et al.* Texture analysis imaging "what a clinical radiologist needs to know". *Eur J Radiol* 2022;**146**:110055.

22. Gatta R, Depeursinge A, Ratib O, *et al*. Integrating radiomics into holomics for personalised oncology: from algorithms to bedside. *Eur Radiol* 2020;**4**(1):1–9.
23. Fornacon-Wood I, Mistry H, Ackermann CJ, *et al*. Reliability and prognostic value of radiomic features are highly dependent on choice of feature extraction platform. *Eur Radiol* 2020;**30**(11):6241–50.
24. Rogers W, Thulasi Seetha S, Refaee TA, *et al*. Radiomics: from qualitative to quantitative imaging. *Br J Radiol* 2020;**93**(1108):20190948.
25. Mali SA, Ibrahim A, Woodruff HC, *et al*. Making radiomics more reproducible across scanner and imaging protocol variations: a review of harmonization methods. *J Pers Med* 2021;**11**(9):842.
26. Yip SS, Aerts HJ. Applications and limitations of radiomics. *PHys Med Biol* 2016;**61**(13):R150.
27. Davnall F, Yip CS, Ljungqvist G, *et al*. Assessment of tumor heterogeneity: an emerging imaging tool for clinical practice? *Insights Imaging* 2012;**3**(6):573–89.
28. Court LE, Fave X, Mackin D, *et al*. Computational resources for radiomics. *Transl Cancer Res* 2016;**5**(4):340–8.
29. Image IBSI. *features—a set of quantitative image features together with the reference values established by the Image Biomarker Standardisation Initiative*. 2022 https://ibsi.readthedocs.io/en/latest/03_Image_features.html. [Accessed 19 July 2022].
30. Ganeshan B, Miles KA. Quantifying tumour heterogeneity with CT. *Cancer Imaging* 2013;**13**(1):140.
31. Liang Z-G, Tan HQ, Zhang F, *et al*. Comparison of radiomics tools for image analyses and clinical prediction in nasopharyngeal carcinoma. *Br J Radiol* 2019;**92**(1102):20190271.
32. Fournier L, Costaridou L, Bidaut L, *et al*. Incorporating radiomics into clinical trials: expert consensus endorsed by the European Society of Radiology on considerations for data-driven compared to biologically driven quantitative biomarkers. *Eur Radiol* 2021;**31**(8):6001–12.
33. Sullivan DC, Obuchowski NA, Kessler LG, *et al*. Metrology standards for quantitative imaging biomarkers. *Radiology* 2015;**277**(3):813–25.
34. van Timmeren JE, Leijenaar RT, van Elmpt W, *et al*. Feature selection methodology for longitudinal cone-beam CT radiomics. *Acta Oncol* 2017;**56**(11):1537–43.
35. Wagner MW, Namdar K, Biswas A, *et al*. Radiomics, machine learning, and artificial intelligence—what the neuroradiologist needs to know. *Neuroradiology* 2021;**63**(12):1957–67.
36. Vial A, Stirling D, Field M, *et al*. The role of deep learning and radiomic feature extraction in cancer-specific predictive modelling: a review. *Transl Cancer Res* 2018;**7**(3):803–16.
37. Avanzo M, Wei L, Stancanello J, *et al*. Machine and deep learning methods for radiomics. *Med Phys* 2020;**47**(5):e185–202.
38. Demircioğlu A. Measuring the bias of incorrect application of feature selection when using cross-validation in radiomics. *Insights Imaging* 2021;**12**(1):1–10.
39. Ramspek CL, Jager KJ, Dekker FW, *et al*. External validation of prognostic models: what, why, how, when and where? *Clin Kidney J* 2020;**14**(1):49–58.
40. Tomaszewski MR, Gillies RJ. The biological meaning of radiomic features. *Radiology* 2021;**298**(3):505.
41. Kim DW, Jang HY, Kim KW, *et al*. Design characteristics of studies reporting the performance of artificial intelligence algorithms for diagnostic analysis of medical images: results from recently published papers. *Korean J Radiol* 2019;**20**(3):405–10.
42. Lambin P, Leijenaar RT, Deist TM, *et al*. Radiomics: the bridge between medical imaging and personalized medicine. *Nat Rev Clin Oncol* 2017;**14**(12):749–62.
43. Ursprung S, Beer L, Bruining A, *et al*. Radiomics of computed tomography and magnetic resonance imaging in renal cell carcinoma—a systematic review and meta-analysis. *Eur Radiol* 2020;**30**(6):3558–66.
44. Mongan J, Moy L, Kahn Jr CE. Checklist for artificial intelligence in medical imaging (CLAIM): a guide for authors and reviewers. *Radiol Artif Intelligence* 2020;**2**(2):e200029.
45. Roberts M, Driggs D, Thorpe M, *et al*. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nat Machine Intell* 2021;**3**(3):199–217.
46. Collins GS, Dhiman P, Navarro CLA, *et al*. Protocol for development of a reporting guideline (TRIPOD-AI) and risk of bias tool (PROBAST-AI) for diagnostic and prognostic prediction model studies based on artificial intelligence. *BMJ Open* 2021;**11**(7):e048008.
47. Limkin E, Sun R, Dercle L, *et al*. Promises and challenges for the implementation of computational medical imaging (radiomics) in oncology. *Ann Oncol* 2017;**28**(6):1191–206.
48. Wu J, Mayer AT, Li R. Integrated imaging and molecular analysis to decipher tumor microenvironment in the era of immunotherapy. *Semi Cancer Biol* 2022;**84**:310–28.
49. Park EK, Lee K-S, Seo BK, *et al*. Machine learning approaches to radiogenomics of breast cancer using low-dose perfusion computed tomography: predicting prognostic biomarkers and molecular subtypes. *Sci Rep* 2019;**9**:1–11.
50. Yan M, Wang W. Development of a radiomics prediction model for histological type diagnosis in solitary pulmonary nodules: the combination of CT and FDG PET. *Front Oncol* 2020;**10**:1–6.
51. Li H, Gao L, Ma H, *et al*. Radiomics-based features for prediction of histological subtypes in central lung cancer. *Front Oncol* 2021;11(April:1–7.
52. Khodabakhshi Z, Mostafaei S, Arabi H, *et al*. Non-small cell lung carcinoma histopathological subtype phenotyping using high-dimensional multinomial multiclass CT radiomics signature. *Comput Biol Med* 2021;**136**:104752.
53. Banerjee S, Wang DS, Kim HJ, *et al*. A computed tomography radiogenomic biomarker predicts microvascular invasion and clinical outcomes in hepatocellular carcinoma. *Hepatology* 2015;**62**:792–800.
54. Renzulli M, Brocchi S, Cucchetti A, *et al*. Can current preoperative imaging be used to detect microvascular invasion of hepatocellular carcinoma? *Radiology* 2016;**279**(2):432–42.
55. Xu X, Zhang H-L, Liu Q-P, *et al*. Radiomic analysis of contrast-enhanced CT predicts microvascular invasion and outcome in hepatocellular carcinoma. *J Hepatol* 2019;**70**(6):1133–44.
56. Qi Y, Zhao T, Han M. The application of radiomics in predicting gene mutations in cancer. *Eur Radiol* 2022;**32**:4014–24.
57. Wan T, Bloch BN, Plecha D, *et al*. A radio-genomics approach for identifying high risk estrogen receptor-positive breast cancers on DCE-MRI: preliminary results in predicting onco type DX risk scores. *Sci Rep* 2016;**6**(October 2015):1–11.
58. Sutton EJ, Oh JH, Dashevsky BZ, *et al*. Breast cancer subtype intertumor heterogeneity: MRI-based features predict results of a genomic assay. *J Magn Reson Imagingg* 2015;**42**:1398–406.
59. Macyszyn L, Akbari H, Pisapia JM, *et al*. Imaging patterns predict patient survival and molecular subtype in glioblastoma via machine learning techniques. *Neuro Oncol* 2016;**18**:417–25.
60. Dasgupta A, Gupta T, Pungavkar S, *et al*. Neuro-oncology magnetic resonance imaging for prediction of molecular subgrouping in medulloblastoma: results from a radiogenomics study of 111 patients. *Neuro Oncol* 2019;**21**(May 2018):115–24.
61. Le NQK, Hung TNK, Do DT, *et al*. Radiomics-based machine learning model for efficiently classifying transcriptome subtypes in glioblastoma patients from MRI. *Comput Biol Med* 2021;**132**:104320.
62. Segal E, Sirlin CB, Ooi C, *et al*. Decoding global gene expression programs in liver cancer by noninvasive imaging. *Nat Biotechnol* 2007;**25**(6):675–80.
63. Gevaert O, Xu J, Hoang CD, *et al*. Non-small cell lung cancer: identifying prognostic imaging biomarkers by leveraging public gene expression microarray data - methods and preliminary results. *Radiology* 2012;**264**(2):387–96.
64. Vargas HA, Huang EP, Lakhman Y, *et al*. Radiogenomics of high-grade serous ovarian cancer: multireader multi-institutional study from the cancer genome atlas ovarian cancer imaging research group. *Radiology* 2017;**285**(2):482–92.
65. Li Z-C, Bai H, Sun Q, *et al*. Multiregional radiomics features from multiparametric MRI for prediction of MGMT methylation status in glioblastoma multiforme: a multicentre study. *Eur Radiol* 2018;**28**:3640–50.
66. Wei J, Yang G, Hao X, *et al*. A multi-sequence and habitat-based MRI radiomics signature for preoperative prediction of MGMT promoter

methylation in astrocytomas with prognostic implication. *Eur Radiol* 2019;**29**:877–88.
67. Vils A, Bogowicz M, Tanadini-Lang S, *et al.* Radiomic analysis to predict outcome in recurrent glioblastoma based on multi-center MR imaging from the prospective DIRECTOR trial. *Front Oncol* 2021;**11**:1–9.
68. Ye F, Hu Y, Gao J, *et al.* Radiogenomics map reveals the landscape of m6A methylation modification pattern in bladder cancer data acquisition of bladder. *Front Immunol* 2021;**12**:1–13.
69. Wang Y, Zheng X-d, Zhu G-Q, *et al.* Crosstalk between metabolism and immune activity reveals four subtypes with therapeutic implications in clear cell renal cell carcinoma. *Front Immunol* 2022;**13**:1–17.
70. Huang C, Cintra M, Brennan K, *et al.* Development and validation of radiomic signatures of head and neck squamous cell carcinoma molecular features and subtypes. *EBioMedicine* 2019;**45**:70–80.
71. Gooden MJM, Bock DGH, Leffers N, *et al.* The prognostic influence of tumour-infiltrating lymphocytes in cancer: a systematic review with meta-analysis. *Br J Cancer* 2011;**19**:93–103.
72. Paijens ST, Vledder A, Bruyn DM, *et al.* Tumor-infiltrating lymphocytes in the immunotherapy era. *Cell Mol Immunol* 2021;**18**(vember 2020):842–59.
73. Yu H, Meng X, Chen H, *et al.* Correlation between mammographic radiomics features and the level of tumor-infiltrating lymphocytes in patients with triple-negative breast cancer. *Front Oncol* 2020;**10**:1–9.
74. Arefan D, Hausler RM, Sumkin JH, *et al.* Predicting cell invasion in breast tumor microenvironment from radiological imaging phenotypes. *BMC Cancer* 2021;**21**:1–9.
75. Xu N, Zhou J, He X, *et al.* Radiomics model for evaluating the level of tumor-infiltrating lymphocytes in breast cancer based on dynamic contrast-enhanced MRI. *Clin Breast Cancer* 2021;**21**(5):440–9.
76. Yu H, Meng X, Chen H, *et al.* Predicting the level of tumor-infiltrating lymphocytes in patients with breast cancer: usefulness of mammographic radiomics features. *Front Oncol* 2021;**11**:1–9.
77. Liao H, Zhang Z, Chen J, *et al.* Preoperative radiomic approach to evaluate tumor-infiltrating CD8+ T cells in hepatocellular carcinoma patients using contrast-enhanced computed tomography. *Ann Surg Oncol* 2019;**26**(13):4537–47.
78. Li J, Shi Z, Liu F, *et al.* XGBoost classifier based on computed tomography radiomics for prediction of tumor-infiltrating CD8+ T-cells in patients with pancreatic ductal adenocarcinoma. *Front Oncol* 2021;11(May:1–12.
79. Tong H, Sun J, Fang J, *et al.* A machine learning model based on PET/CT radiomics and clinical characteristics predicts tumor immune profiles in non-small cell lung cancer: a retrospective multicohort study. *Front Immunol* 2022;13(April:1–11.
80. Wen Q, Yang Z, Zhu J, *et al.* Pretreatment CT-based radiomics signature as a potential imaging biomarker for predicting the expression of PD-L1 and CD8 + TILs in ESCC. *OncoTargets Ther* 2020;**13**:12003–13.
81. Hyuck S, Jin Y, Koh J, *et al.* A radiomic signature model to predict the chemoradiation-induced alteration in tumor-infiltrating CD8+ cells in locally advanced rectal cancer. *Radiother Oncol* 2021;**162**:124–31.
82. Sun R, Limkin EJ, Vakalopoulou M, *et al.* A radiomics approach to assess tumour-infiltrating CD8 cells and response to anti-PD-1 or anti-PD-L1 immunotherapy: an imaging biomarker, retrospective multicohort study. *Lancet Oncol* 2018;**19**(9):1180–91.
83. Veeraraghavan H, Friedman CF, Delair DF, *et al.* Machine learning-based prediction of microsatellite instability and high tumor mutation burden from contrast - enhanced computed tomography in endometrial cancers. *Sci Rep* 2020:1–10.
84. Hoshino I, Yokota H, Iwatate Y, *et al.* Prediction of the differences in tumor mutation burden between primary and metastatic lesions by radiogenomics. *Cancer Sci* 2021;**113**:229–39.
85. Golia JS, Johan P, Chakraborty J, *et al.* Radiomics-based prediction of microsatellite instability in colorectal cancer at initial computed tomography evaluation. *Abdom Radiol* 2019;**44**(11):3755–63.
86. Huang Z, Zhang W, He D, *et al.* Development and validation of a radiomics model based on T2WI images for preoperative prediction of microsatellite instability status in rectal cancer. *Medicine* 2020;**99**:6–10.
87. Liang X, Wu Y, Liu Y, *et al.* A multicenter study on the preoperative prediction of gastric cancer microsatellite instability status based on computed tomography radiomics. *Abdom Radiol* 2022;**47**(6):2036–45.
88. Tang C, Hobbs B, Amer A, *et al.* Development of an immune-pathology informed radiomics model for non-small cell lung cancer. *Sci Rep* 2018;**8**(1):1–9.
89. Zhou J, Zou S, Kuang D, *et al.* A novel approach using FDG-PET/CT-based radiomics to assess tumor immune phenotypes in patients with non-small cell lung cancer. *Front Oncol* 2021;**11**:1–12.
90. Aoude LG, Wong BZY, Bonazzi VF, *et al.* Radiomics biomarkers correlate with CD8 expression and predict immune signatures in melanoma patients. *Mol Cancer Res* 2019;(10):950–6.
91. Ren J, Tian J, Yuan Y, *et al.* Magnetic resonance imaging based radiomics signature for the preoperative discrimination of stage I–II and III–IV head and neck squamous cell carcinoma. *Eur J Radiol* 2018;**106**:1–6.
92. Aerts HJ, Velazquez ER, Leijenaar RT, *et al.* Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach. *Nat Commun* 2014;**5**(1):1–9.
93. Coroller TP, Grossmann P, Hou Y, *et al.* CT-based radiomic signature predicts distant metastasis in lung adenocarcinoma. *Radiother Oncol* 2015;**114**(3):345–50.
94. Zhou H, Dong D, Chen B, *et al.* Diagnosis of distant metastasis of lung cancer: based on clinical and radiomic features. *Translat Oncol* 2018;**11**(1):31–6.
95. Wu J, Aguilera T, Shultz D, *et al.* Early-stage non-small cell lung cancer: quantitative imaging characteristics of 18F fluorodeoxyglucose PET/CT allow prediction of distant metastasis. *Radiology* 2016;**281**(1):270.
96. Liu S, Zhang Y, Chen L, *et al.* Whole-lesion apparent diffusion coefficient histogram analysis: significance in T and N staging of gastric cancers. *BMC Cancer* 2017;**17**(1):1–9.
97. Liu S, Zhang Y, Xia J, *et al.* Predicting the nodal status in gastric cancers: the role of apparent diffusion coefficient histogram characteristic analysis. *Magn Reson Imaging* 2017;**42**:144–51.
98. Liu S, Zheng H, Zhang Y, *et al.* Whole-volume apparent diffusion coefficient-based entropy parameters for assessment of gastric cancer aggressiveness. *J Magn Reson Imaging* 2018;**47**(1):168–75.
99. Granzier RWY, Nijnatten VTJA, Woodru HC, *et al.* Exploring breast cancer response prediction to neoadjuvant systemic therapy using MRI-based radiomics: a systematic review. *Eur J Radiol* 2019;**121**(July).
100. Nie K, Shi L, Chen Q, *et al.* Rectal cancer: assessment of neoadjuvant chemoradiation outcome based on radiomics of multiparametric MRI. *Clin Cancer Res* 2016;**22**(15):5256–64.
101. Liu Z, Zhang X-Y, Shi Y-J, *et al.* Radiomics analysis for evaluation of pathological complete response to neoadjuvant chemoradiotherapy in locally advanced rectal cancer. *Clin Cancer Res* 2017;**23**(16):7253–62.
102. Wan L, Zhang C, Zhao Q, *et al.* Developing a prediction model based on MRI for pathological complete response after neoadjuvant chemoradiotherapy in locally advanced rectal cancer. *Abdom Radiol* 2019;**44**(9):2978–87.
103. Zhou X, Yi Y, Liu Z, *et al.* Radiomics-based pretherapeutic prediction of non-response to neoadjuvant therapy in locally advanced rectal cancer. *Ann Surg Oncol* 2019;**26**(6):1676–84.
104. Shaish H, Aukerman A, Vanguri R, *et al.* Radiomics of MRI for pretreatment prediction of pathologic complete response, tumor regression grade, and neoadjuvant rectal score in patients with locally advanced rectal cancer undergoing neoadjuvant chemoradiation: an international multicenter study. *Eur Radiol* 2020;**30**:6263–73.
105. Sun C, Tian X, Liu Z, *et al.* Radiomic analysis for pretreatment prediction of response to neoadjuvant chemotherapy in locally advanced cervical cancer: a multicentre study. *EBioMedicine* 2019;**46**:160–9.
106. Zhao L, Gong J, Xi Y, *et al.* MRI-based radiomics nomogram may predict the response to induction chemotherapy and survival in locally advanced nasopharyngeal carcinoma. *Eur Radiol* 2020;**30**:537–46.
107. Wang W, Peng Y, Feng X, *et al.* Development and validation of a computed tomography-based radiomics signature to predict response to neoadjuvant chemotherapy for locally advanced gastric cancer. *JAMA Netw Open* 2021;**4**(8):1–14.
108. Cha KH, Hadjiiski L, Chan H-P, *et al.* Bladder cancer treatment response assessment in CT using radiomics with deep-learning. *Sci Rep* 2017;**7**:1–12.
109. Khorrami M, Khunger M, Zagouras A, *et al.* Combination of peri- and intratumoral radiomic features on baseline CT scans predicts response to chemotherapy in lung adenocarcinoma. *Radiol Artif Intell* 2019;**1**(2).

110. Vaidya P, Bera K, Gupta A, *et al.* CT derived radiomic score for predicting the added benefit of adjuvant chemotherapy following surgery in stage I, II resectable non-small cell lung cancer: a retrospective multicohort study for outcome prediction. *Lancet Digital Health* 2020;**2**:116–28.
111. Rundo L, Beer L, Sanchez LE, *et al.* Clinically interpretable radiomics-based prediction of histopathologic response to neoadjuvant chemotherapy in high-grade serous ovarian carcinoma. *Front Oncol* 2022;**12**.
112. Nougaret S, McCague C, Tibermacine H, *et al.* Radiomics and radiogenomics in ovarian cancer: a literature review. *Abdom Radiol* 2020;**46**:2308–22.
113. Hu Y, Xie C, Yang H, *et al.* Assessment of intratumoral and peritumoral computed tomography radiomics for predicting pathological complete response to neoadjuvant chemoradiation in patients with esophageal squamous cell carcinoma. *JAMA Netw Open* 2020;**3**(9):1–14.
114. Khorrami M, Jain P, Bera K, *et al.* Predicting pathologic response to neoadjuvant chemoradiation in resectable stage III non-small cell lung cancer patients using computed tomography radiomic features. *Lung Cancer* 2020;**135**:1–9.
115. Xu Y, Hosny A, Zeleznik R, *et al.* Deep learning predicts lung cancer treatment response from serial medical imaging. *Clin Cancer Res* 2019;**25**(11):3266–75.
116. Aerts HJWL, Grossmann P, Tan Y, *et al.* Defining a radiomic response phenotype: a pilot study using targeted therapy in NSCLC. *Sci Rep* 2016;**6**:33860.
117. Song J, Shi J, Dong D, *et al.* A new approach to predict progression-free survival in stage IV EGFR-mutant NSCLC patients with EGFR-TKI therapy. *Clin Cancer Res* 2018;**24**(15):3583–92.
118. Song J, Wang L, Ng NN, *et al.* Development and validation of a machine learning model to explore tyrosine kinase inhibitor response in patients with stage IV EGFR variant-positive non-small cell lung cancer. *JAMA Netw Open* 2020;**3**(12):1–13.
119. Braman N, Prasanna P, Whitney J, *et al.* Association of peritumoral radiomics with tumor biology and pathologic response to preoperative targeted therapy for HER2 (ERBB2)-Positive Breast Cancer. *JAMA Netw Open* 2019;**2**:1–18.
120. Bitencourt VAG, Gibbs P, Rossi C, *et al.* MRI-based machine learning radiomics can predict HER2 expression level and pathologic response after neoadjuvant therapy in HER2 overexpressing breast cancer. *EBioMedicine* 2020;**61**:103042.
121. Kong C, Zhao Z, Chen W, *et al.* Prediction of tumor response via a pretreatment MRI radiomics-based nomogram in HCC treated with TACE. *Eur Radiol* 2021;**31**:7500–11.
122. Meng X-P, Wang Y-C, Ju S, *et al.* Radiomics analysis on multiphase contrast-enhanced CT: a survival prediction tool in patients with hepatocellular carcinoma undergoing transarterial chemoembolization. *Front Oncol* 2020;**10**:1–12.
123. Trebeschi S, Drago SG, Birkbak NJ, *et al.* Predicting response to cancer immunotherapy using noninvasive radiomic biomarkers. *Ann Oncol* 2019;**30**:998–1004.
124. Wu M, Zhang Y, Zhang J, *et al.* A combined-radiomics approach of CT images to predict response to anti-PD-1 immunotherapy in NSCLC: a retrospective multicenter study. *Front Oncol* 2022;**11**:1–13.
125. Kim B-C, Kim J, Kim K, *et al.* Preliminary radiogenomic evidence for the prediction of metastasis and chemotherapy response in pediatric patients. *Cancers* 2021;**13**:1–11.
126. Fan M, Cui Y, You MSC, *et al.* Radiogenomic signatures of oncotype DX recurrence score enable prediction of survival in estrogen receptor-positive breast cancer: a multicohort study. *Radiology* 2022;**302**(3):516–24.
127. Jimenez JE, Abdelhafez A, Mittendorf EA, *et al.* A model combining pretreatment MRI radiomic features and tumor-infiltrating lymphocytes to predict response to neoadjuvant systemic therapy in triple-negative breast cancer. *Eur J Radiol* 2022;**149**:110220.
128. Nardone V, Reginelli A, Grassi R, *et al.* Delta radiomics: a systematic review. *Radiol Med* 2021:1–13.
129. Gong J, Bao X, Wang T, *et al.* A short-term follow-up CT based radiomics approach to predict response to immunotherapy in advanced non-small-cell lung cancer. *Oncoimmunology* 2022;**11**(1):2028962.
130. Barabino E, Rossi G, Pamparino S, *et al.* Exploring response to immunotherapy in non-small cell lung cancer using delta-radiomics. *Cancers* 2022;**14**(2):350.
131. Guerrisi A, Russillo M, Loi E, *et al.* Exploring CT texture parameters as predictive and response imaging biomarkers of survival in patients with metastatic melanoma treated with PD-1 inhibitor nivolumab: a pilot study using a delta-radiomics approach. *Front Oncol* 2021;**11**:704607.
132. Liu Y, Wu M, Zhang Y, *et al.* Imaging biomarkers to predict and evaluate the effectiveness of immunotherapy in advanced non-small-cell lung cancer. *Front Oncol* 2021;**11**:657615.
133. Lee G, Bak SH, Lee HY, *et al.* Measurement variability in treatment response determination for non-small cell lung cancer. *J Thorac Imaging* 2019;**34**(2):103–15.
134. Wang ZL, Mao LL, Zhou ZG, *et al.* Pilot study of CT-based radiomics model for early evaluation of response to immunotherapy in patients with metastatic melanoma. *Front Oncol* 2020;**10**:1524.
135. Basler L, Gabryś HS, Hogan SA, *et al.* Radiomics, tumor volume, and blood biomarkers for early prediction of pseudoprogression in patients with metastatic melanoma treated with immune checkpoint inhibition radiomics and biomarkers for prediction of pseudoprogression. *Clin Cancer Res* 2020;**26**(16):4414–25.
136. Yu F, Hang J, Deng J, *et al.* Radiomics features on ultrasound imaging for the prediction of diseasefree survival in triple negative breast cancer: a multi-institutional study. *Br J Radiol* 2021;**94**:20210188.
137. Kickingereder P, Burth S, Wick A, *et al.* Radiomic profiling of glioblastoma: identifying an imaging predictor of patient survival with improved performance over established clinical and radiologic risk models. *Radiology* 2016;**280**(3):880–9.
138. Bae S, Choi YS, Ahn SS, *et al.* Radiomic MRI phenotyping of glioblastoma: improving survival prediction. *Radiology* 2018;**289**:797–806.
139. Choi Y, Nam Y, Jang J, *et al.* Radiomics may increase the prognostic value for survival in glioblastoma patients when combined with conventional clinical and genetic prognostic models. *Eur Radiol* 2021;**31**(4):2084–93.
140. Choi Y, Ahn KJ, Nam Y, *et al.* Analysis of heterogeneity of peritumoral T2 hyperintensity in patients with pretreatment glioblastoma: prognostic value of MRI-based radiomics. *Eur J Radiol* 2019;**120**:108642.
141. Kickingereder P, Neuberger U, Bonekamp D, *et al.* Radiomic subtyping improves disease stratification beyond key molecular, clinical, and standard imaging characteristics in patients with glioblastoma. *Neuro Oncol* 2018;**20**(6):848–57.
142. Sushentsev N, Rundo L, Blyuss O, *et al.* MRI-derived radiomics model for baseline prediction of prostate cancer progression on active surveillance. *Sci Rep* 2021;**11**:1–11.
143. Cen C, Liu L, Li X, *et al.* Pancreatic ductal adenocarcinoma at CT: a combined nomogram model to preoperatively predict cancer stage and survival outcome. *Front Oncol* 2021;**11**:1–12.
144. Grossmann P, Stringfield O, El-Hachem N, *et al.* Defining the biological basis of radiomic phenotypes in lung cancer. *eLife* 2017;**6**:1–22.
145. Huang Y, Zeng H, Chen L, *et al.* Exploration of an integrative prognostic model of radiogenomics features with underlying gene expression patterns in clear cell renal cell carcinoma. *Front Oncol* 2021;**11**:1–14.
146. Wijethilake N, Islam M, Ren H. Radiogenomics model for overall survival prediction of glioblastoma. *Med Biol Eng Comput* 2020;**58**(8):1767–77.
147. Vlachavas E-I, Pilalis E, Papadodima O, *et al.* Radiogenomic analysis of F-18-fluorodeoxyglucose positron emission tomography and gene expression data elucidates the epidemiological complexity of colorectal cancer landscape. *Comput Struct Biotechnol J* 2019;**17**:177–85.
148. Eary JF. Cancer imaging program update: 2020. *Radiol Imaging Cancer* 2020;**2**(4).
149. McAteer M, O'Connor JP, Koh D, *et al.* Introduction to the National Cancer Imaging Translational Accelerator (NCITA): a UK-wide infrastructure for multicentre clinical translation of cancer imaging biomarkers. *Br J Cancer* 2021;**125**(11):1462–5.
150. Sollini M, Antunovic L, Chiti A, *et al.* Towards clinical application of image mining: a systematic review on artificial intelligence and radiomics. *Eur J Nucl Med Mol Imaging* 2019;**46**(13):2656–72.

151. Oliveira M, Fernandes P, Avelar W, *et al*. Texture analysis of computed tomography images of acute ischemic stroke patients. *Brazil J Med Biol Res* 2009;**42**:1076–9.
152. Peter R, Korfiatis P, Blezek D, *et al*. A quantitative symmetry-based analysis of hyperacute ischemic stroke lesions in noncontrast computed tomography. *Med Phys* 2017;**44**(1):192–9.
153. Alwalid O, Long X, Xie M, *et al*. CT angiography-based radiomics for classification of intracranial aneurysm rupture. *Front Neurol* 2021;**12**:619864.
154. Zhang Y, Zhang B, Liang F, *et al*. Radiomics features on non-contrast-enhanced CT scan can precisely classify AVM-related hematomas from other spontaneous intraparenchymal hematoma types. *Eur Radiol* 2019;**29**(4):2157–65.
155. Wang H, Sun Y, Ge Y, *et al*. A clinical–radiomics nomogram for functional outcome predictions in ischemic stroke. *Neurol Ther* 2021;**10**(2):819–32.
156. Kanazawa T, Takahashi S, Minami Y, *et al*. Early prediction of clinical outcomes in patients with aneurysmal subarachnoid hemorrhage using computed tomography texture analysis. *J Clin Neurosci* 2020;**71**:144–9.
157. Cui H, Wang X, Bian Y, *et al*. Ischemic stroke clinical outcome prediction based on image signature selection from multimodality data. *Annu Int Conf IEEE Eng Med Biol Soc* 2018;**2018**:722–5.
158. Tang TY, Jiao Y, Cui Y, *et al*. Penumbra-based radiomics signature as prognostic biomarkers for thrombolysis of acute ischemic stroke patients: a multicenter cohort study. *J Neurol* 2020;**267**(5):1454–63.
159. Betrouni N, Yasmina M, Bombois S, *et al*. Texture features of magnetic resonance images: an early marker of post-stroke cognitive impairment. *Transl Stroke Res* 2020;**11**(4):643–52.
160. Kassner A, Liu F, Thornhill RE, *et al*. Prediction of hemorrhagic transformation in acute ischemic stroke using texture analysis of postcontrast T1-weighted MR images. *J Magn Reson Imaging* 2009;**30**(5):933–41.
161. Shen Q, Shan Y, Hu Z, *et al*. Quantitative parameters of CT texture analysis as potential markersfor early prediction of spontaneous intracranial hemorrhage enlargement. *Eur Radiol* 2018;**28**(10):4389–96.
162. Ma C, Zhang Y, Niyazi T, *et al*. Radiomics for predicting hematoma expansion in patients with hypertensive intraparenchymal hematomas. *Eur J Radiol* 2019;**115**:10–5.
163. Li H, Xie Y, Wang X, *et al*. Radiomics features on non-contrast computed tomography predict early enlargement of spontaneous intracerebral hemorrhage. *Clin Neurol Neurosurg* 2019;**185**:105491.
164. Xie H, Ma S, Wang X, *et al*. Noncontrast computer tomography-based radiomics model for predicting intracerebral hemorrhage expansion: preliminary findings and comparison with conventional radiological model. *Eur Radiol* 2020;**30**(1):87–98.
165. Fu B, Qi S, Tao L, *et al*. Image patch-based net water uptake and radiomics models predict malignant cerebral edema after ischemic stroke. *Front Neurol* 2020;**11**:609747.
166. Jiang L, Zhang C, Wang S, *et al*. MRI radiomics features from infarction and cerebrospinal fluid for prediction of cerebral edema after acute ischemic stroke. *Front Aging Neurosci* 2022;**14**:782036.
167. Yao X, Liao L, Han Y, *et al*. Computerized tomography radiomics features analysis for evaluation of perihematomal edema in basal ganglia hemorrhage. *J Craniofac Surg* 2019;**30**(8):e768–71.
168. Qiu W, Kuang H, Nair J, *et al*. Radiomics-based intracranial thrombus features on CT and CTA predict recanalization with intravenous alteplase in patients with acute ischemic stroke. *AJNR Am J Neuroradiol* 2019;**40**(1):39–44.
169. Sarioglu O, Sarioglu FC, Capar AE, *et al*. Clot-based radiomics features predict first pass effect in acute ischemic stroke. *Interv Neuroradiol* 2022;**28**(2):160–8.
170. Hofmeister J, Bernava G, Rosi A, *et al*. Clot-based radiomics predict a mechanical thrombectomy strategy for successful recanalization in acute ischemic stroke. *Stroke* 2020;**51**(8):2488–94.
171. Mathias JM, Tofts PS, Losseff NA. Texture analysis of spinal cord pathology in multiple sclerosis. *Magn Reson Med* 1999;**42**(5):929–35.
172. Zhang Y, Moore GRW, Laule C, *et al*. Pathological correlates of magnetic resonance imaging texture heterogeneity in multiple sclerosis. *Ann Neurol* 2013;**74**(1):91–9.
173. Peng Y, Zheng Y, Tan Z, *et al*. Prediction of unenhanced lesion evolution in multiple sclerosis using radiomics-based models: a machine learning approach. *Mult Scler Relat Disord* 2021;**53**:102989.
174. Yu O, Mauss Y, Zollner G, *et al*. Distinct patterns of active and non-active plaques using texture analysis on brain NMR images in multiple sclerosis patients: preliminary results. *Magn Reson Imaging* 1999;**17**(9):1261–7.
175. Caruana G, Pessini LM, Cannella R, *et al*. Texture analysis in susceptibility-weighted imaging may be useful to differentiate acute from chronic multiple sclerosis lesions. *Eur Radiol* 2020;**30**(11):6348–56.
176. Lavrova E, Lommers E, Woodruff HC, *et al*. Exploratory radiomic analysis of conventional vs. quantitative brain MRI: toward automatic diagnosis of early multiple sclerosis. *Front Neurosci* 2021;**15**:679941.
177. Cao X, Wang X, Xue C, *et al*. A radiomics approach to predicting Parkinson's disease by incorporating whole-brain functional activity and gray matter structure. *Front Neurosci* 2020;**14**.
178. Shi D, Zhang H, Wang G, *et al*. Machine learning for detecting Parkinson's disease by resting-state functional magnetic resonance imaging: a multicenter radiomics analysis. *Front Aging Neurosci* 2022;**14**:806828.
179. Shi D, Yao X, Li Y, *et al*. Classification of Parkinson's disease using a region-of-interest- and resting-state functional magnetic resonance imaging-based radiomics approach. *Brain Imaging Behav* 2022;**16**:2150–63.
180. Rahmim A, Huang P, Shenkov N, *et al*. Improved prediction of outcome in Parkinson's disease using radiomics analysis of longitudinal DAT SPECT images. *NeuroImage: Clin* 2017;**16**:539–44.
181. Salmanpour MR, Shamsaei M, Saberi A, *et al*. Robust identification of Parkinson's disease subtypes using radiomics and hybrid machine learning. *Comput Biol Med* 2021;**129**:104142.
182. Pang H, Yu Z, Li R, *et al*. MRI-based radiomics of basal nuclei in differentiating idiopathic Parkinson's disease from parkinsonian variants of multiple system atrophy: a susceptibility-weighted imaging study. *Front Aging Neurosci* 2020;**12**.
183. Feng Q, Ding Z. MRI radiomics classification and prediction in Alzheimer's disease and mild cognitive impairment: a review. *Curr Alzheimer Res* 2020;**17**(3):297–309.
184. Zhang J, Yu C, Jiang G, *et al*. 3D texture analysis on MRI images of Alzheimer's disease. *Brain Imaging Behav* 2012;**6**(1):61–9.
185. Bang M, Eom J, An C, *et al*. An interpretable multiparametric radiomics model for the diagnosis of schizophrenia using magnetic resonance imaging of the corpus callosum. *Translat Psychiatr* 2021;**11**(1):1–8.
186. Park YW, Choi D, Lee J, *et al*. Differentiating patients with schizophrenia from healthy controls by hippocampal subfields using radiomics. *Schizophrenia Res* 2020;**223**:337–44.
187. Cui LB, Fu YF, Liu L, *et al*. Baseline structural and functional magnetic resonance imaging predicts early treatment response in schizophrenia with radiomics strategy. *Eur J Neurosci* 2021;**53**(6):1961–75.
188. Kolossváry M, Karády J, Szilveszter B, *et al*. Radiomic features are superior to conventional quantitative computed tomographic metrics to identify coronary plaques with napkin-ring sign. *Circ Cardiovasc Imaging* 2017;**10**(12):e006843.
189. Kolossváry M, Karády J, Kikuchi Y, *et al*. Radiomics versus visual and histogram-based assessment to identify atheromatous lesions at coronary CT angiography: an ex vivo study. *Radiology* 2019;**293**(1):89–96.
190. Lin A, Kolossváry M, Cadet S, *et al*. Radiomics-based precision phenotyping identifies unstable coronary plaques from computed tomography angiography. *JACC Cardiovasc Imaging* 2022;**15**(5):859–71.
191. Kolossváry M, Gerstenblith G, Bluemke DA, *et al*. Contribution of risk factors to the development of coronary atherosclerosis as confirmed via coronary CT angiography: a longitudinal radiomics-based study. *Radiology* 2021;**299**(1):97–106.
192. Schoepf UJ, Emrich T. A brave new world: toward precision phenotyping and understanding of coronary artery disease using radiomics plaque analysis. *Radiology* 2021;**299**(1):107–8.
193. Spadarella G, Perillo T, Ugga L, *et al*. Radiomics in cardiovascular disease imaging: from pixels to the heart of the problem. *Curr Cardiovasc Imaging Rep* 2022;**15**(2):11–21.
194. Hinzpeter R, Wagner MW, Wurnig MC, *et al*. Texture analysis of acute myocardial infarction with CT: first experience study. *PLoS One* 2017;**12**(11):e0186876.
195. Benjamins JW, Yeung MW, Maaniitty T, *et al*. Improving patient identification for advanced cardiac imaging through machine learning-integration of clinical and coronary CT angiography data. *Int J Cardiol* 2021;**335**:130–6.

196. Di Noto T, von Spiczak J, Mannil M, *et al*. Radiomics for distinguishing myocardial infarction from myocarditis at late gadolinium enhancement at MRI: comparison with subjective visual analysis. *Radiol Cardiothorac Imaging* 2019;**1**(5):e180026.
197. Si N, Shi K, Li N, *et al*. Identification of patients with acute myocardial infarction based on coronary CT angiography: the value of pericoronary adipose tissue radiomics. *Eur Radiol* 2022;**32**(10):6868–77.
198. Peng Y, Wu K, Wang YXJ, *et al*. Association between cine CMR-based radiomics signature and microvascular obstruction in patients with ST-segment elevation myocardial infarction. *J Thorac Dis* 2022;**14**(4):969–78.
199. Gibbs T, Villa ADM, Sammut E, *et al*. Quantitative assessment of myocardial scar heterogeneity using cardiovascular magnetic resonance texture analysis to risk stratify patients post-myocardial infarction. *Clin Radiol* 2018;**73**(12).
200. Kotu LP, Engan K, Borhani R, *et al*. Cardiac magnetic resonance image-based classification of the risk of arrhythmias in post-myocardial infarction patients. *Artif Intell Med* 2015;**64**(3):205–15.
201. Antunes S, Esposito A, Palmisanov A, *et al*. *Characterization of normal and scarred myocardium based on texture analysis of cardiac computed tomography images. 38th annual international conference of the IEEE engineering in medicine and biology society (EMBC) 2016*. Piscataway: IEEE; 2016.
202. Larroza A, López-Lereu MP, Monmeneu JV, *et al*. Texture analysis of cardiac cine magnetic resonance imaging to detect nonviable segments in patients with chronic myocardial infarction. *Med Phys* 2018;**45**(4):1471–80.
203. Baessler B, Mannil M, Oebel S, *et al*. Subacute and chronic left ventricular myocardial scar: accuracy of texture analysis on nonenhanced cine MR images. *Radiology* 2018;**286**(1):103–12.
204. Baessler B, Luecke C, Lurz J, *et al*. Cardiac MRI and texture analysis of myocardial T1 and T2 maps in myocarditis with acute versus chronic symptoms of heart failure. *Radiology* 2019;**292**(3):608–17.
205. Blagova O, Osipova Y, Nedostup A, *et al*. Diagnostic value of different noninvasive criteria of latent myocarditis in comparison with myocardial biopsy. *Cardiology* 2019;**142**(3):167–74.
206. Schofield R, Ganeshan B, Fontana M, *et al*. Texture analysis of cardiovascular magnetic resonance cine images differentiates aetiologies of left ventricular hypertrophy. *Clin Radiol* 2019;**74**(2):140–9.
207. Cheng S, Fang M, Cui C, *et al*. LGE-CMR-derived texture features reflect poor prognosis in hypertrophic cardiomyopathy patients with systolic dysfunction: preliminary results. *Eur Radiol* 2018;**28**(11):4615–24.
208. Amano Y, Suzuki Y, Yanagisawa F, *et al*. Relationship between extension or texture features of late gadolinium enhancement and ventricular tachyarrhythmias in hypertrophic cardiomyopathy. *BioMed Res Int* 2018;**2018**.
209. Chan RH, Maron BJ, Olivotto I, *et al*. Prognostic value of quantitative contrast-enhanced cardiovascular magnetic resonance for the evaluation of sudden death risk in patients with hypertrophic cardiomyopathy. *Circulation* 2014;**130**(6):484–95.
210. Kagiyama N, Shrestha S, Cho JS, *et al*. A low-cost texture-based pipeline for predicting myocardial tissue remodeling and fibrosis using cardiac ultrasound. *EBioMedicine* 2020;**54**:102726.
211. Nam K, Suh YJ, Han K, *et al*. Value of computed tomography radiomic features for differentiation of periprosthetic mass in patients with suspected prosthetic valve obstruction. *Circ Cardiovasc Imaging* 2019;**12**(11):e009496.
212. Kotze CW, Rudd JH, Ganeshan B, *et al*. CT signal heterogeneity of abdominal aortic aneurysm as a possible predictive biomarker for expansion. *Atherosclerosis* 2014;**233**(2):510–7.
213. Wang Y, Zhou M, Ding Y, *et al*. A radiomics model for predicting the outcome of endovascular abdominal aortic aneurysm repair based on machine learning. *Vascular* 2022:17085381221091061.
214. Ding N, Hao Y, Wang Z, *et al*. CT texture analysis predicts abdominal aortic aneurysm post-endovascular aortic aneurysm repair progression. *Sci Rep* 2020;**10**(1):1–10.
215. Araki T, Jain PK, Suri HS, *et al*. Stroke risk stratification and its validation using ultrasonic echolucent carotid wall plaque morphology: a machine learning paradigm. *Comp Biol Med* 2017;**80**:77–96.
216. Acharya RU, Faust O, Alvin APC, *et al*. Symptomatic vs. asymptomatic plaque classification in carotid ultrasound. *J Med Sys* 2012;**36**(3):1861–71.
217. Wu G, Yang P, Xie Y, *et al*. Development of a clinical decision support system for severity risk prediction and triage of COVID-19 patients at hospital admission: an international multicentre study. *Eur Respir J* 2020;**56**(2).
218. Wang L, Kelly B, Lee EH, *et al*. Multi-classifier-based identification of COVID-19 from chest computed tomography using generalizable and interpretable radiomics features. *Eur J Radiol* 2021;**136**:109552.
219. Wang H, Wang L, Lee EH, *et al*. Decoding COVID-19 pneumonia: comparison of deep learning and radiomics CT image signatures. *Eur J Nucl Med Mol Imaging* 2021;**48**(5):1478–86.
220. Chen Y, Wang Y, Zhang Y, *et al*. A quantitative and radiomics approach to monitoring ARDS in COVID-19 patients based on chest CT: a retrospective cohort study. *Int J Med Sci* 2020;**17**(12):1773–82.
221. Purkayastha S, Xiao Y, Jiao Z, *et al*. Machine learning-based prediction of COVID-19 severity and progression to critical illness using CT imaging and clinical data. *Korean J Radiol* 2021;**22**(7):1213–24.
222. Li L, Wang L, Zeng F, *et al*. Development and multicenter validation of a CT-based radiomics signature for predicting severe COVID-19 pneumonia. *Eur Radiol* 2021;**31**(10):7901–12.
223. Li C, Dong D, Li L, *et al*. Classification of severe and critical COVID-19 using deep learning and radiomics. *IEEE J Biomed Health Inform* 2020;**24**(12):3585–94.
224. Bae J, Kapse S, Singh G, *et al*. Predicting mechanical ventilation and mortality in COVID-19 using radiomics and deep learning on chest radiographs: a multi-institutional study. *Diagnostics (Basel)* 2021;**11**(10).
225. Wu Q, Wang S, Li L, *et al*. Radiomics analysis of computed tomography helps predict poor prognostic outcome in COVID-19. *Theranostics* 2020;**10**(16):7231–44.
226. Varghese BA, Shin H, Desai B, *et al*. Predicting clinical outcomes in COVID-19 using radiomics on chest radiographs. *Br J Radiol* 2021;**94**(1126):20210221.
227. Xin X, Mo R, Shao M, *et al*. A human–computer collaboration for COVID-19 differentiation: combining a radiomics model with deep learning and human auditing. *Ann Palliat Med* 2021;**10**(7):7329–39.
228. Chen Z, Li X, Li J, *et al*. A COVID-19 risk score combining chest CT radiomics and clinical characteristics to differentiate COVID-19 pneumonia from other viral pneumonias. *Aging (Albany NY)* 2021;**13**(7):9186–224.
229. Koyuncu H, Barstuğan M. COVID-19 discrimination framework for radiograph images by considering radiomics, selective information, feature ranking, and a novel hybrid classifier. *Signal Process Image Commun* 2021;**97**:116359.
230. Kao YS, Lin KT. A meta-analysis of computerized tomography-based radiomics for the diagnosis of COVID-19 and viral pneumonia. *Diagnostics (Basel)* 2021;**11**(6).
231. Gülbay M, Özbay BO, Mendi BAR, *et al*. A CT radiomics analysis of COVID-19-related ground-glass opacities and consolidation: is it valuable in a differential diagnosis with other atypical pneumonias? *PLoS One* 2021;**16**(3):e0246582.
232. Castaldi PJ, San José Estépar R, Mendoza CS, *et al*. Distinct quantitative computed tomography emphysema patterns are associated with physiology and function in smokers. *Am J Respir Crit Care Med* 2013;**188**(9):1083–90.
233. Occhipinti M, Paoletti M, Bartholmai BJ, *et al*. Spirometric assessment of emphysema presence and severity as measured by quantitative CT and CT-based radiomics in COPD. *Resp Res* 2019;**20**(1):1–11.
234. Yun J, Cho YH, Lee SM, *et al*. Deep radiomics-based survival prediction in patients with chronic obstructive pulmonary disease. *Sci Rep* 2021;**11**(1):15144.
235. Li Z, Liu L, Zhang Z, *et al*. A novel CT-based radiomics features analysis for identification and severity staging of COPD. *Acad Radiol* 2022;**29**(5):663–73.
236. Lafata KJ, Zhou Z, Liu JG, *et al*. An exploratory radiomics approach to quantifying pulmonary function in CT images. *Sci Rep* 2019;**9**(1):11509.
237. Yang K, Yang Y, Kang Y, *et al*. The value of radiomic features in chronic obstructive pulmonary disease assessment: a prospective study. *Clin Radiol* 2022;**77**(6):e466–72.

238. Yang Y, Li W, Guo Y, *et al*. Early COPD risk decision for adults aged from 40 to 79 years based on lung radiomics features. *Front Med (Lausanne)* 2022;**9**:845286.
239. Ginsburg SB, Lynch DA, Bowler RP, *et al*. Automated texture-based quantification of centrilobular nodularity and centrilobular emphysema in chest CT images. *Acad Radiol* 2012;**19**(10):1241–51.
240. Refaee T, Wu G, Ibrahim A, *et al*. The emerging role of radiomics in COPD and lung cancer. *Respiration* 2020;**99**(2):99–107.
241. Mets O, De Jong P, Van Ginneken B, *et al*. Quantitative computed tomography in COPD: possibilities and limitations. *Lung* 2012;**190**(2):133–45.
242. Guo F, Capaldi D, Kirby M, *et al*. Development of a pulmonary imaging biomarker pipeline for phenotyping of chronic lung disease. *J Med Imaging (Bellingham)* 2018;**5**(2):026002.
243. Stefano A, Gioè M, Russo G, *et al*. Performance of radiomics features in the quantification of idiopathic pulmonary fibrosis from HRCT. *Diagnostics (Basel).* 2020;**10**(5).
244. Refaee T, Salahuddin Z, Frix AN, *et al*. Diagnosis of idiopathic pulmonary fibrosis in high-resolution computed tomography scans using a combination of handcrafted radiomics and deep learning. *Front Med (Lausanne)* 2022;**9**:915243.
245. Liang CH, Liu YC, Wan YL, *et al*. Quantification of cancer-developing idiopathic pulmonary fibrosis using whole-lung texture analysis of HRCT images. *Cancers (Basel).* 2021;**13**(22).
246. Wang K, Lu X, Zhou H, *et al*. Deep learning radiomics of shear wave elastography significantly improved diagnostic performance for assessing liver fibrosis in chronic hepatitis B: a prospective multicentre study. *Gut* 2019;**68**(4):729–41.
247. Lubner MG, Malecki K, Kloke J, *et al*. Texture analysis of the liver at MDCT for assessing hepatic fibrosis. *Abdom Radiol* 2017;**42**(8):2069–78.
248. Daginawala N, Li B, Buch K, *et al*. Using texture analyses of contrast enhanced CT to assess hepatic fibrosis. *Eur J Radiol* 2016;**85**(3):511–7.
249. Liu F, Ning Z, Liu Y, *et al*. Development and validation of a radiomics signature for clinically significant portal hypertension in cirrhosis (CHESS1701): a prospective multicenter study. *EBioMedicine* 2018;**36**:151–8.
250. Li H, Mo Y, Huang C, *et al*. An MSCT-based radiomics nomogram combined with clinical factors can identify Crohn's disease and ulcerative colitis. *Ann Transl Med* 2021;**9**(7):572.
251. Laterza L, Boldrini L, Tran H, *et al*. P359 Radiomics could predict surgery at, 10 years in Crohn's disease. *J Crohn's Colitis* 2022;**16**(Suppl. 1):i367–8.
252. Chen Y, Li H, Feng J, *et al*. A novel radiomics nomogram for the prediction of secondary loss of response to infliximab in Crohn's disease. *J Inflamm Res* 2021;**14**:2731–40.
253. Panayides AS, Pattichis MS, Leandrou S, *et al*. Radiogenomics for precision medicine with a big data analytics perspective. *IEEE J Biomed Health Inform* 2018;**23**(5):2063–79.
254. Kai C, Uchiyama Y, Shiraishi J, *et al*. Computer-aided diagnosis with radiogenomics: analysis of the relationship between genotype and morphological changes of the brain magnetic resonance images. *Radiol Phys Technol* 2018;**11**(3):265–73.
255. Huang Y, Li L, Jiang J. Radiogenomics of Alzheimer's disease: exploring gene related metabolic imaging markers. *Annu Int Conf IEEE Eng Med Biol Soc* 2021;**2021**:5772–5.
256. Budzikowski JD, Foy JJ, Rashid AA, *et al*. Radiomics-based assessment of idiopathic pulmonary fibrosis is associated with genetic mutations and patient survival. *J Med Imaging (Bellingham)* 2021;**8**(3):031903.
257. Oikonomou EK, Williams MC, Kotanidis CP, *et al*. A novel machine learning-derived radiotranscriptomic signature of perivascular fat improves cardiac risk prediction using coronary CT angiography. *Eur Heart J* 2019;**40**(43):3529–43.
258. Kotanidis CP, Xie C, Adlam D, *et al*. D Radiotranscriptomic analysis of perivascular adipose tissue quantifies vascular inflammation in covid-19 from routine CT angiograms: stratification of "new UK variant" Infection and prediction of in-hospital outcomes. *Heart* 2021;**107**:A177–8.
259. *RCR. Position statement on artificial intelligence*. 2018. Available atl: https://www.rcr.ac.uk/posts/rcr-position-statement-artificial-intelligence. [Accessed 15 July 2022].
260. Lambin P, Rios-Velazquez E, Leijenaar R, *et al*. Radiomics: extracting more information from medical images using advanced feature analysis. *Eur J Cancer* 2012;**48**(4):441–6.
261. Abadi M, Agarwal A, Barham P, et al. Tensorflow: large-scale machine learning on heterogeneous distributed systems. *arXiv 160304467*. 2016.
262. Google- PAIR. PAIR saliency: framework-agnostic implementation for state-of-the-art saliency methods (XRAI, BlurIG, SmoothGrad, and more). https://pair-code.github.io/saliency/#home. [Accessed 15 July 2022].
263. Chowdhury ME, Rahman T, Khandakar A, *et al*. Can AI help in screening viral and COVID-19 pneumonia? *IEEE Access* 2020;**8**:132665–76.
264. Rahman T, Khandakar A, Qiblawey Y, *et al*. Exploring the effect of image enhancement techniques on COVID-19 detection using chest radiograph images. *Comp Biol Med* 2021;**132**:104319.